

20-24 Липня 2026

Data Science School 2026



Юрій Чирка

Head of ML

It-Jim



Вступ до методів класифікації та регресії

День 3: 22.07.26

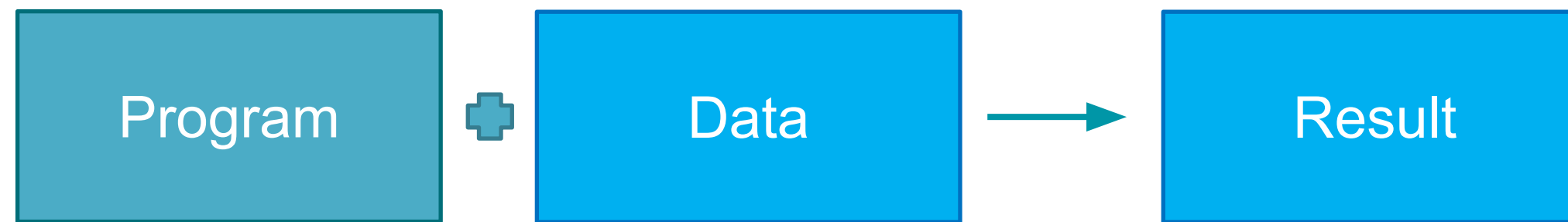
- 01 КОНЦЕПТ МАШИННОГО НАВЧАННЯ**
модель, основні типи задач
- 02 МЕТОДИ КЛАСИФІКАЦІЇ**
kNN, дерева прийняття рішення, ансамблі
- 03 МЕТОДИ РЕГРЕСІЇ**
Аналогія з класифікацією, поліноміальна регресія
- 04 ДЕКІЛЬКА СЛІВ ПРО НЕЙРОННІ МЕРЕЖІ**
Штучний нейрон, найпростіша архітектура

Machine Learning: How Does That Work?

- “Machine learning algorithms build a mathematical model based on sample data, known as “training data”, in order to make predictions or decisions without being explicitly programmed to perform the task” – Wikipedia.

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Normal programming

Machine Learning: How Does That Work?

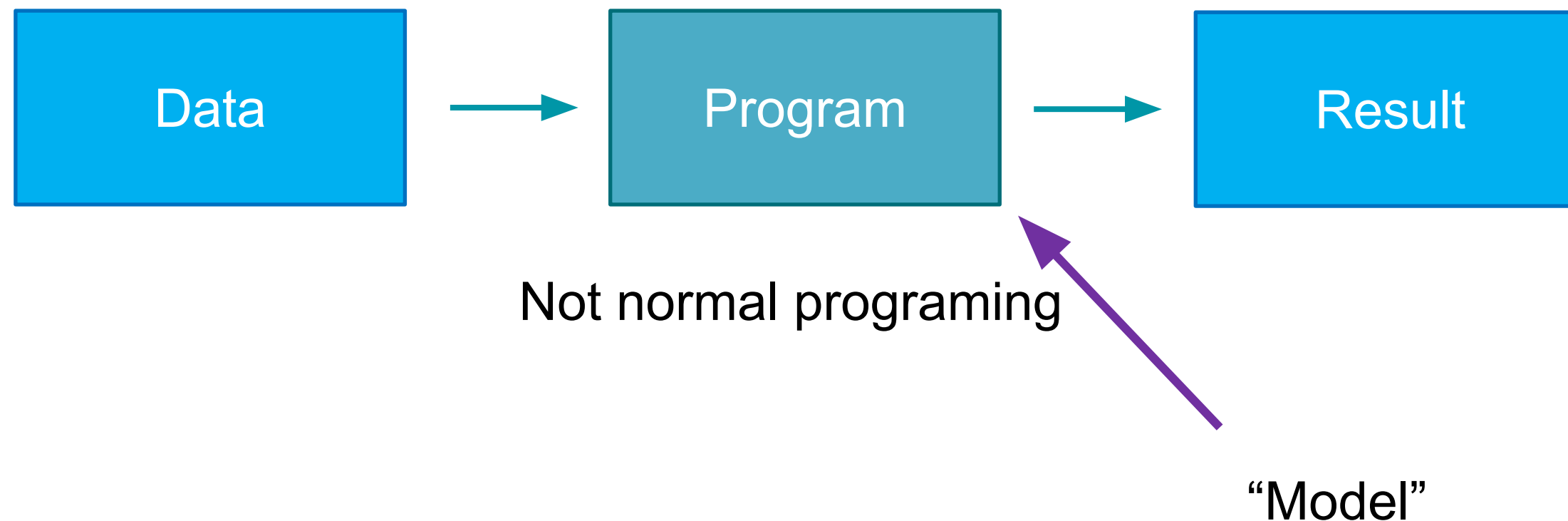
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Not normal programming

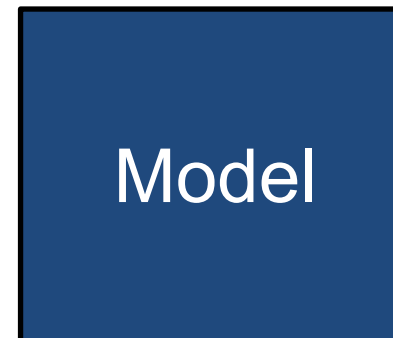
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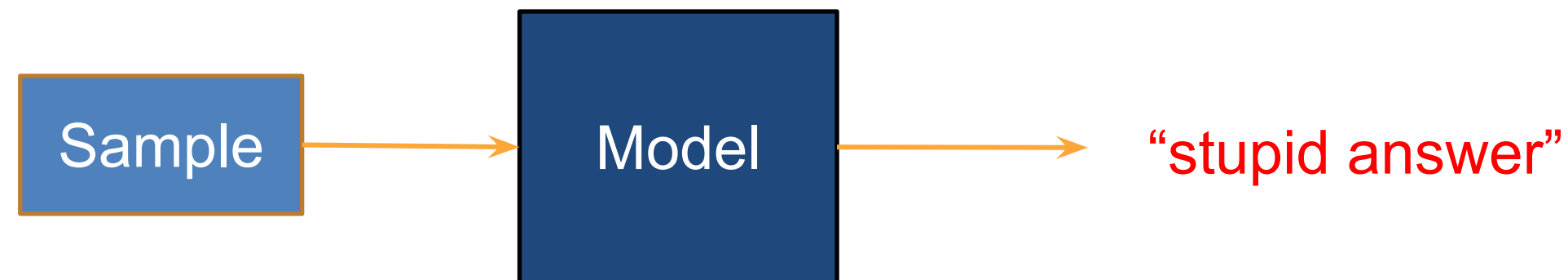
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Framework – no any learning yet.

Machine Learning: How Does That Work?

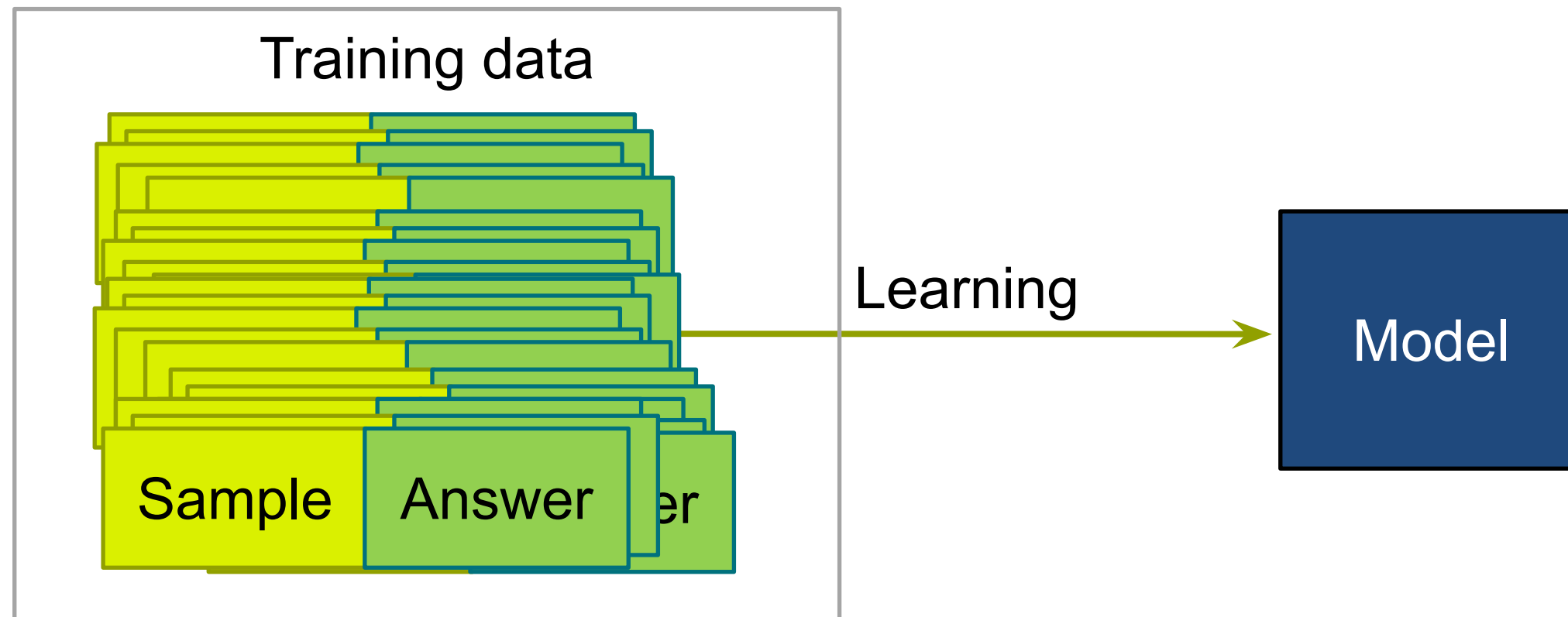
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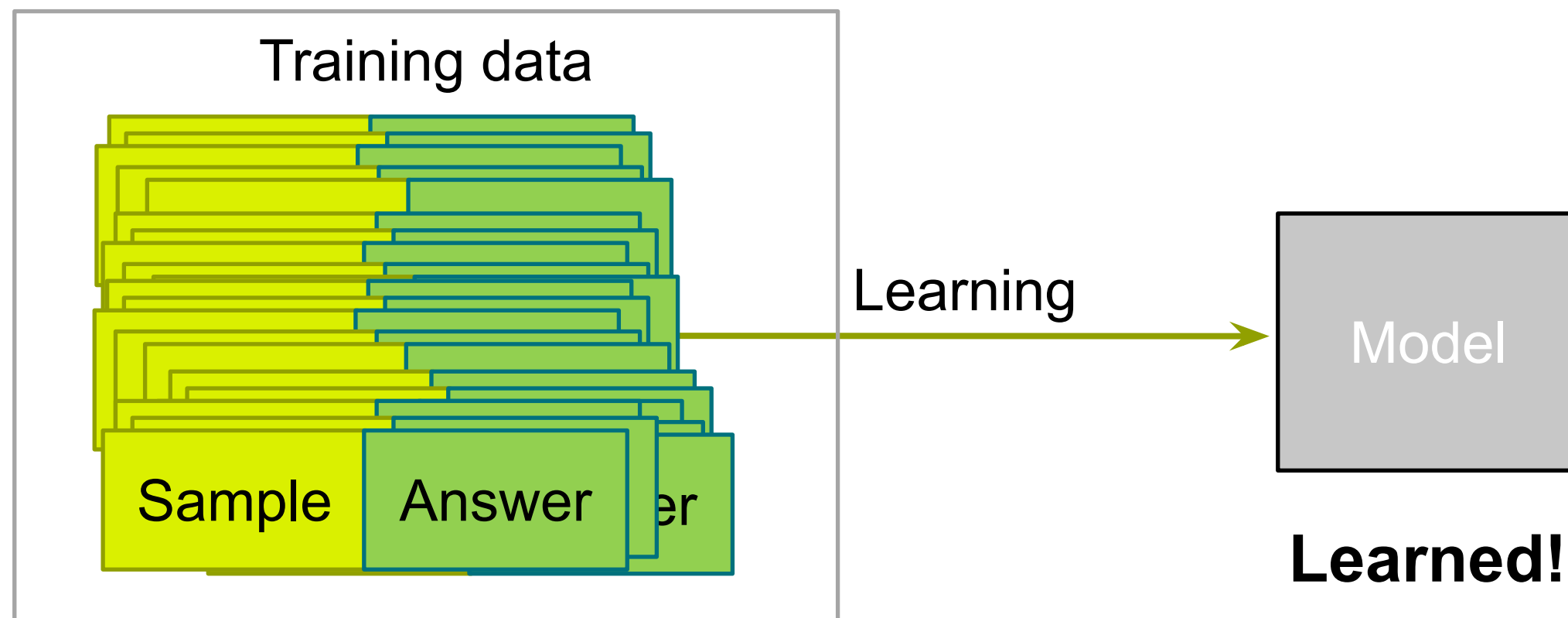
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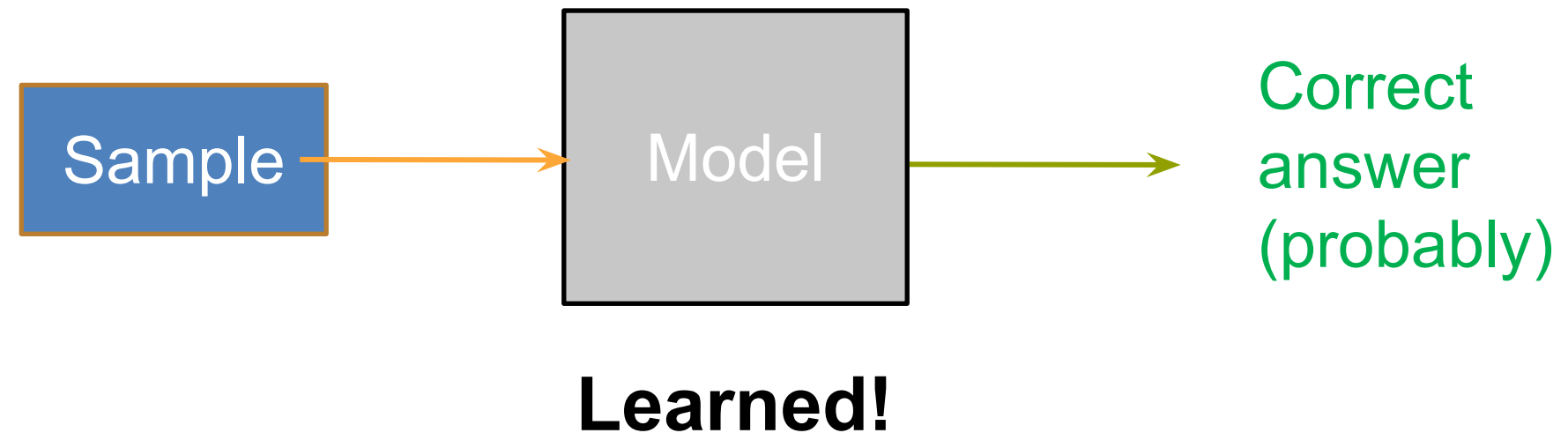
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Supervised ML Types

- Classification**

Area	Bedrooms	Price	Location	Type
30	1	28000	PP	Apartment
150	4	200000	Zh	House
60	3	60000	AI	Apartment

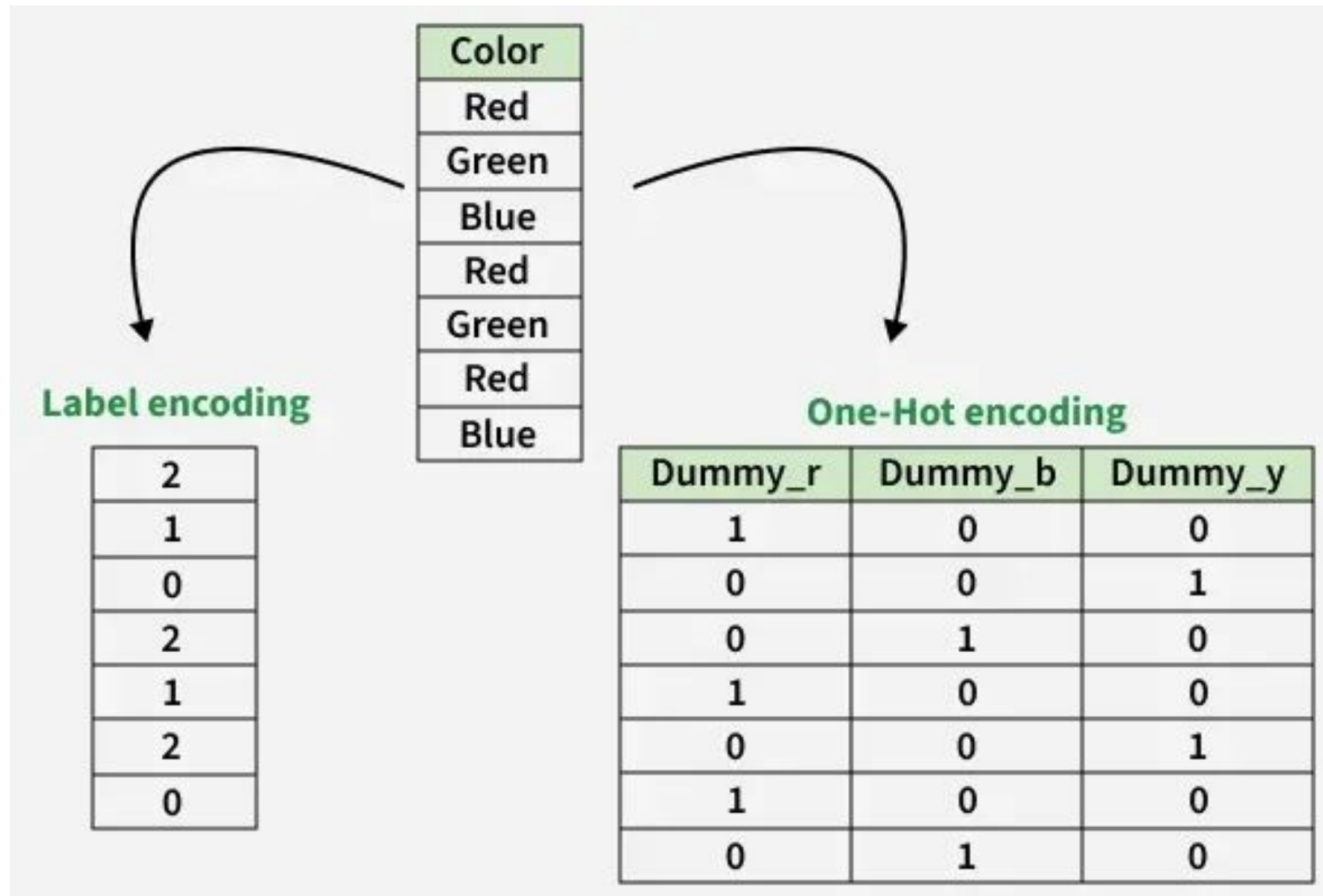
Supervised ML Types

- **Classification**
- **Regression**

Area	Bedrooms	Type	Location	Price
30	1	Apartment	PP	28000
150	4	House	Zh	200000
60	3	Apartment	AI	60000

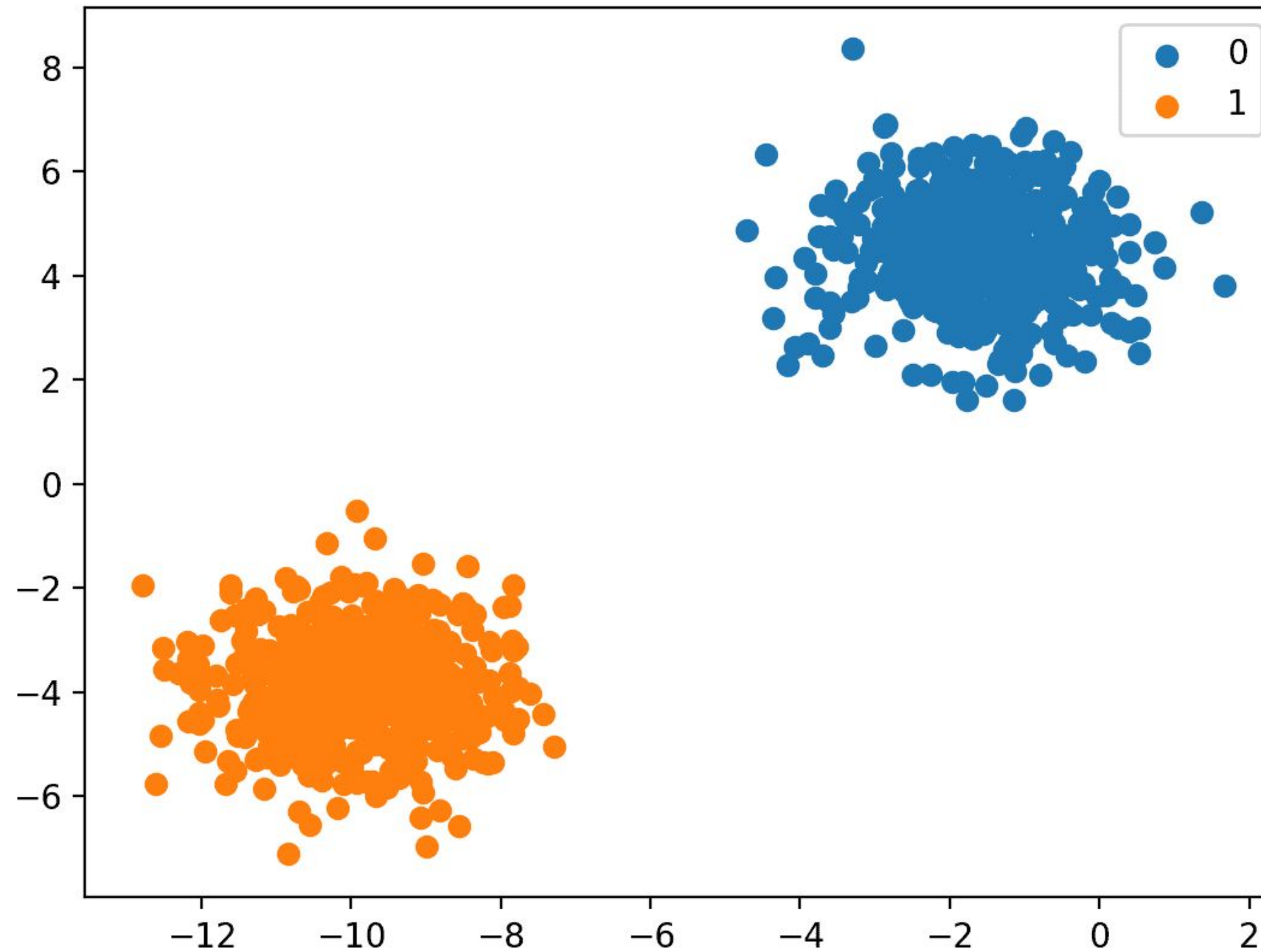
Categorical data processing

One-hot encoding

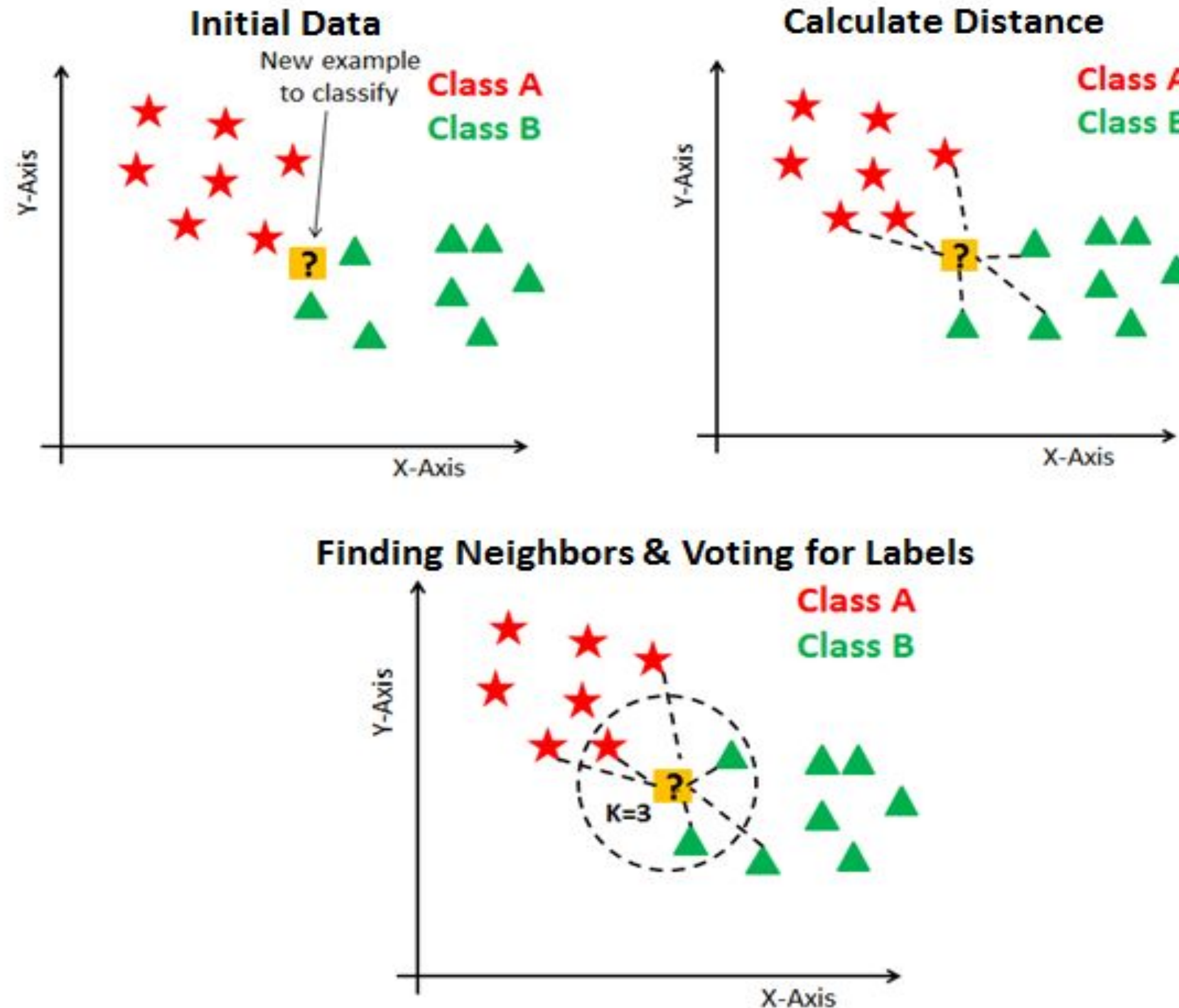


Classification

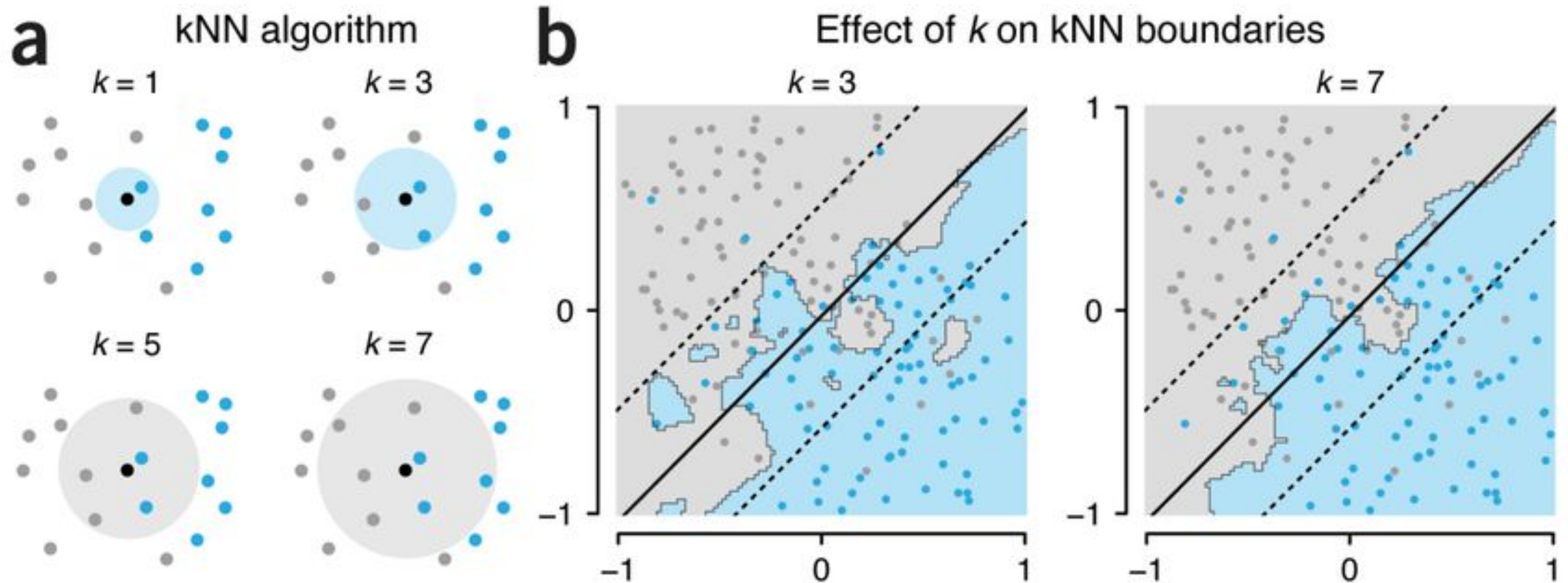
Binary classification



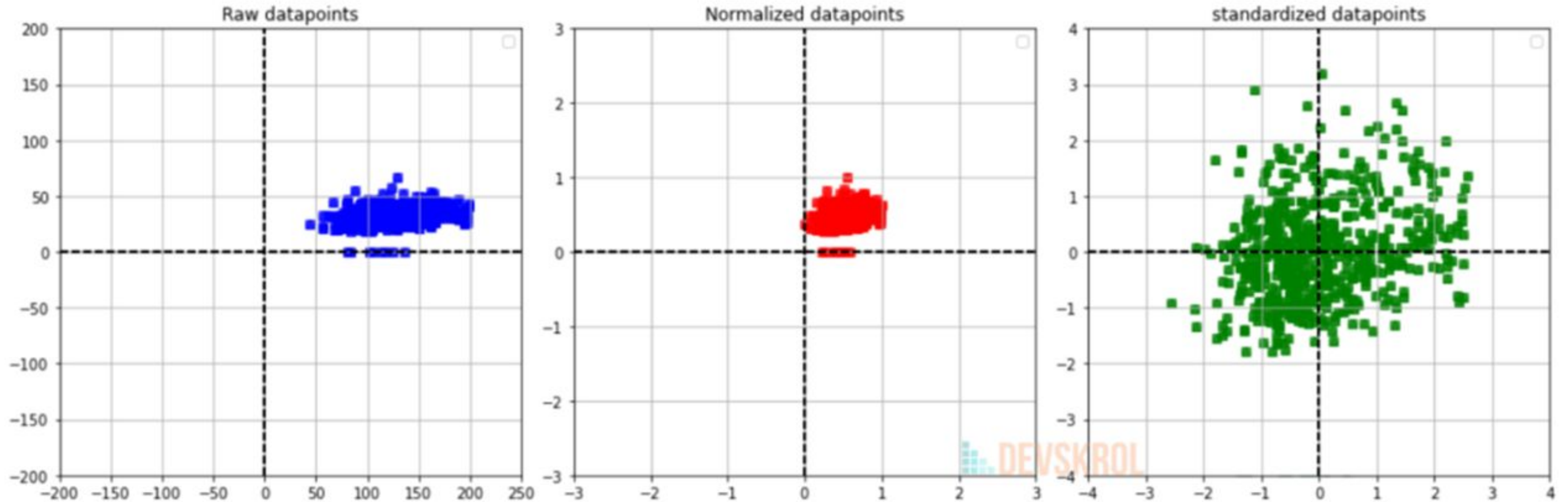
K Nearest Neighbors (kNN)



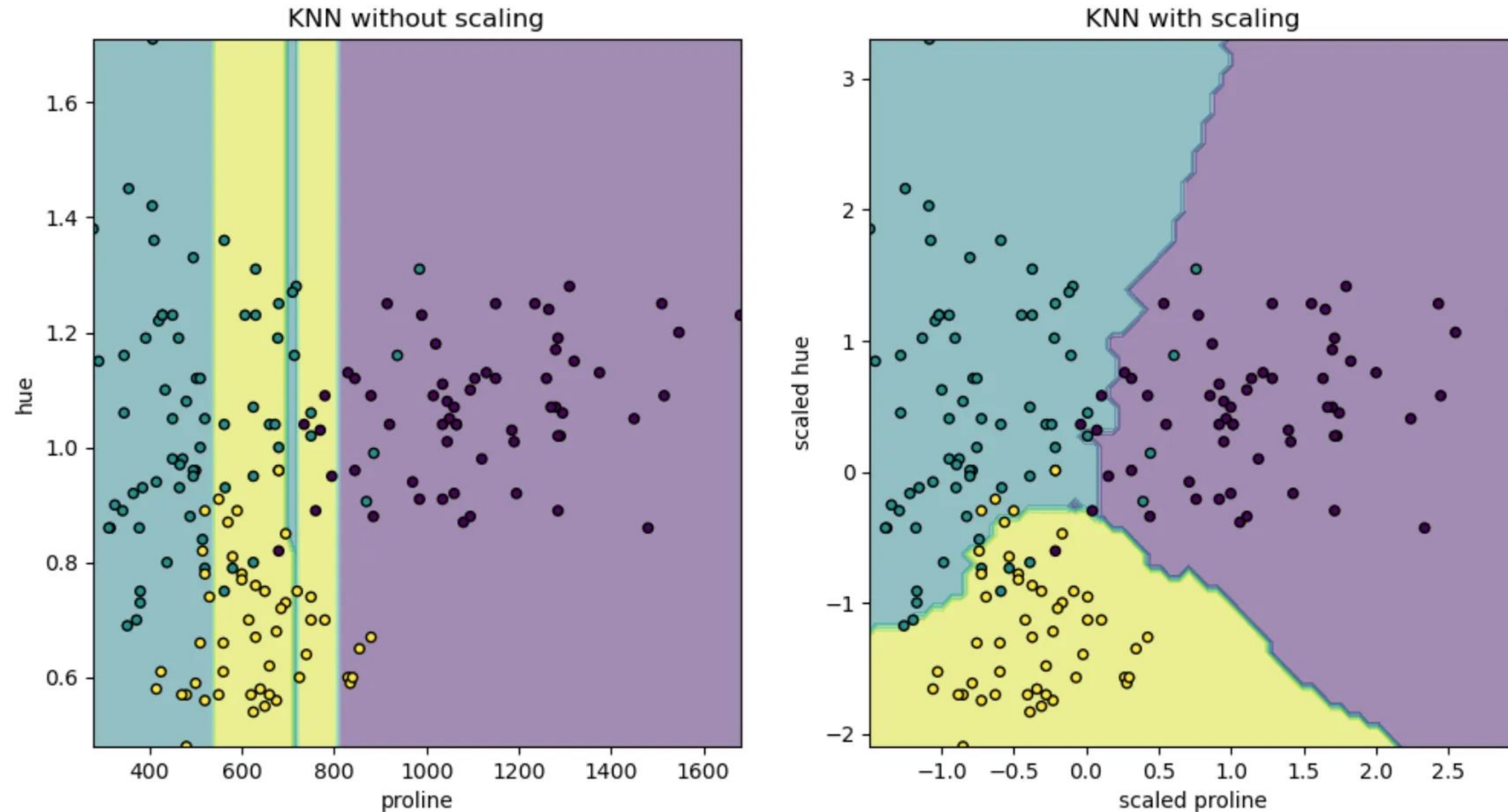
K Nearest Neighbors (kNN)



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K Nearest Neighbors (kNN)



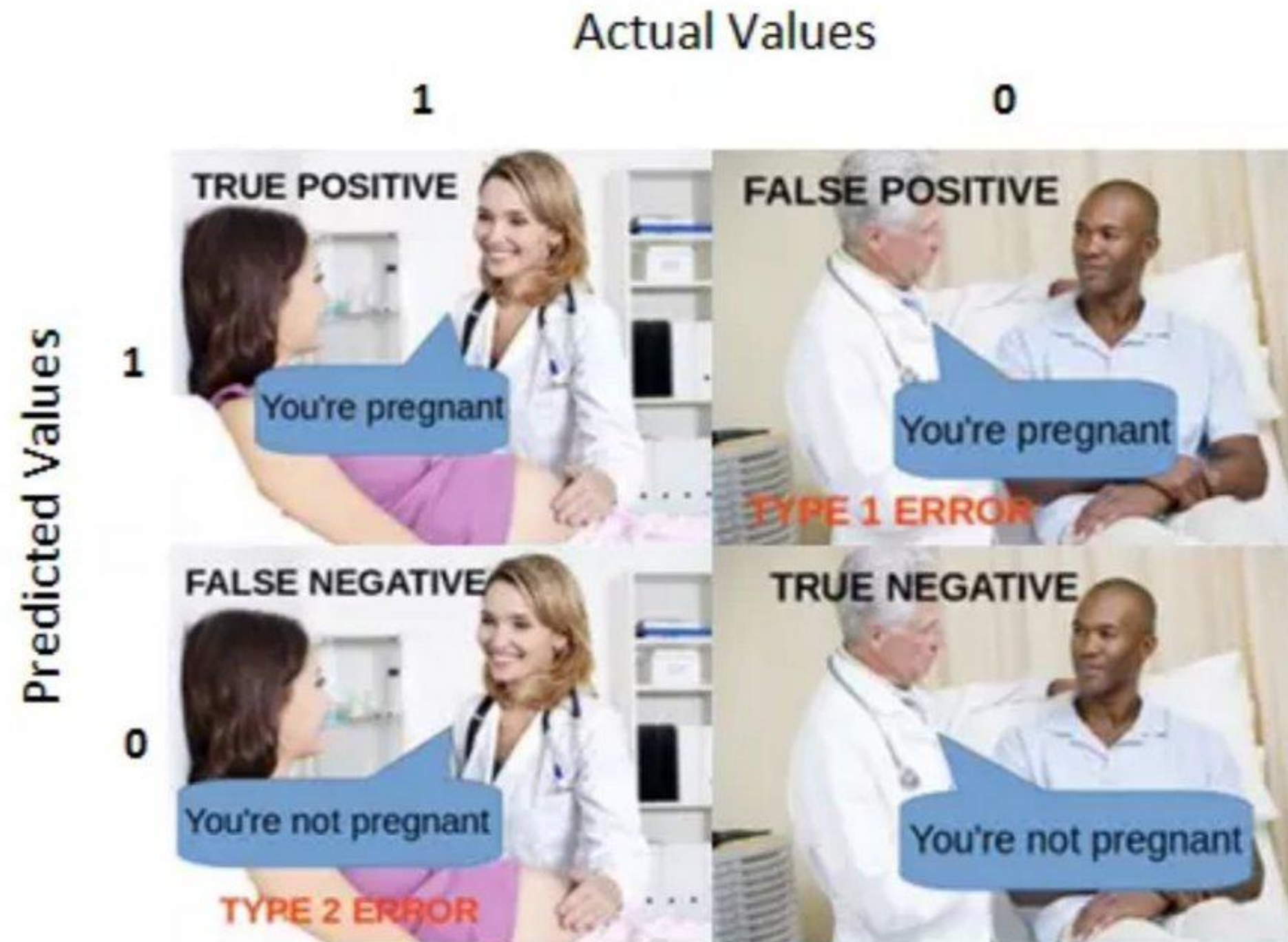
Classification

Binary classification

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Classification

Binary classification



Classification

Binary classification main metrics

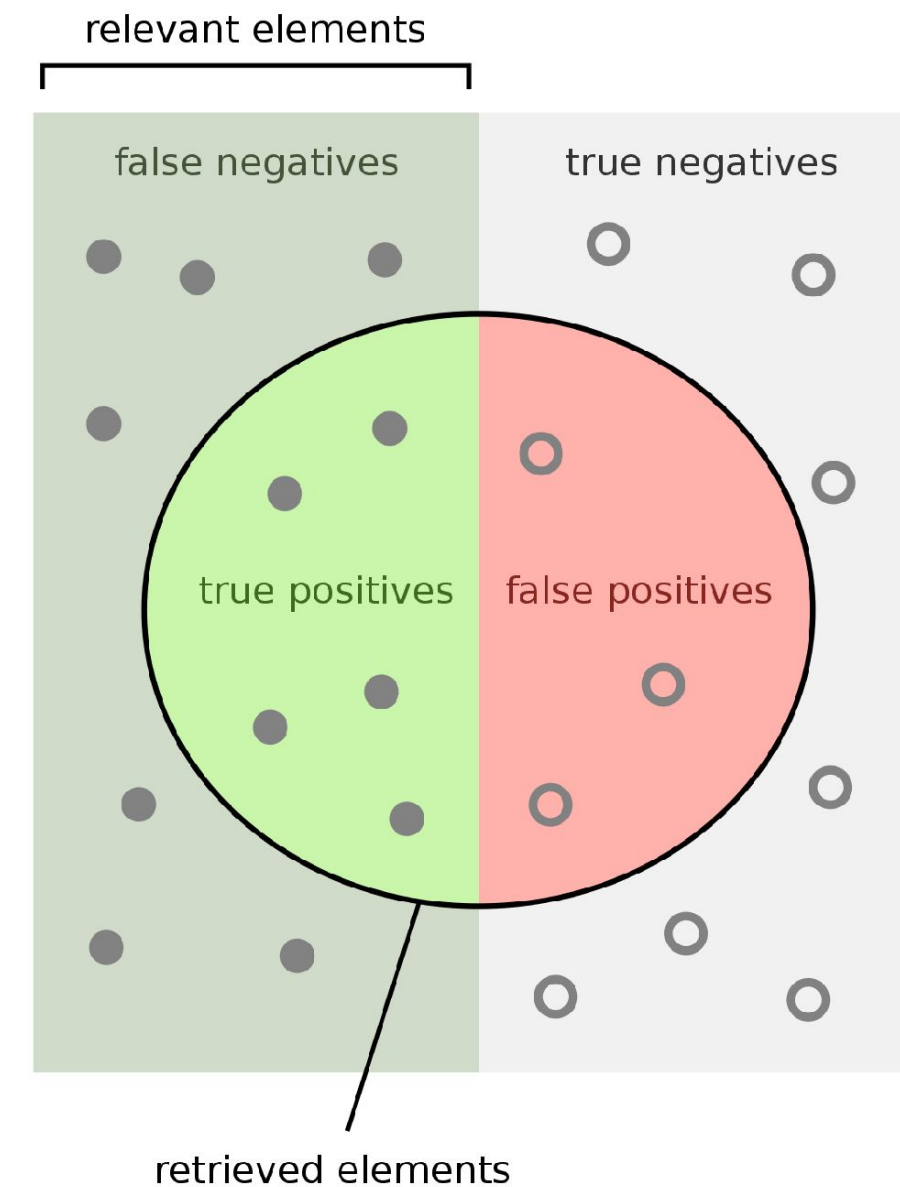
		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

$$\text{Precision} = \frac{tp}{tp + fp}$$

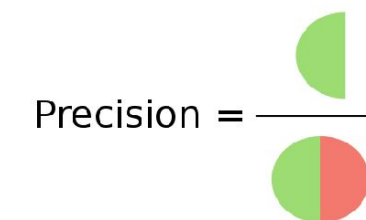
$$\text{Recall} = \frac{tp}{tp + fn}$$

$$F = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

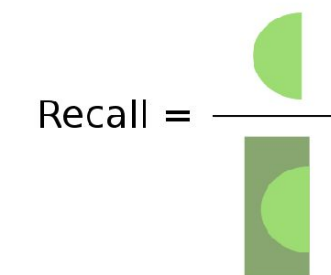
$$\text{Accuracy} = \frac{tp + tn}{tp + tn + fp + fn}$$



How many retrieved items are relevant?



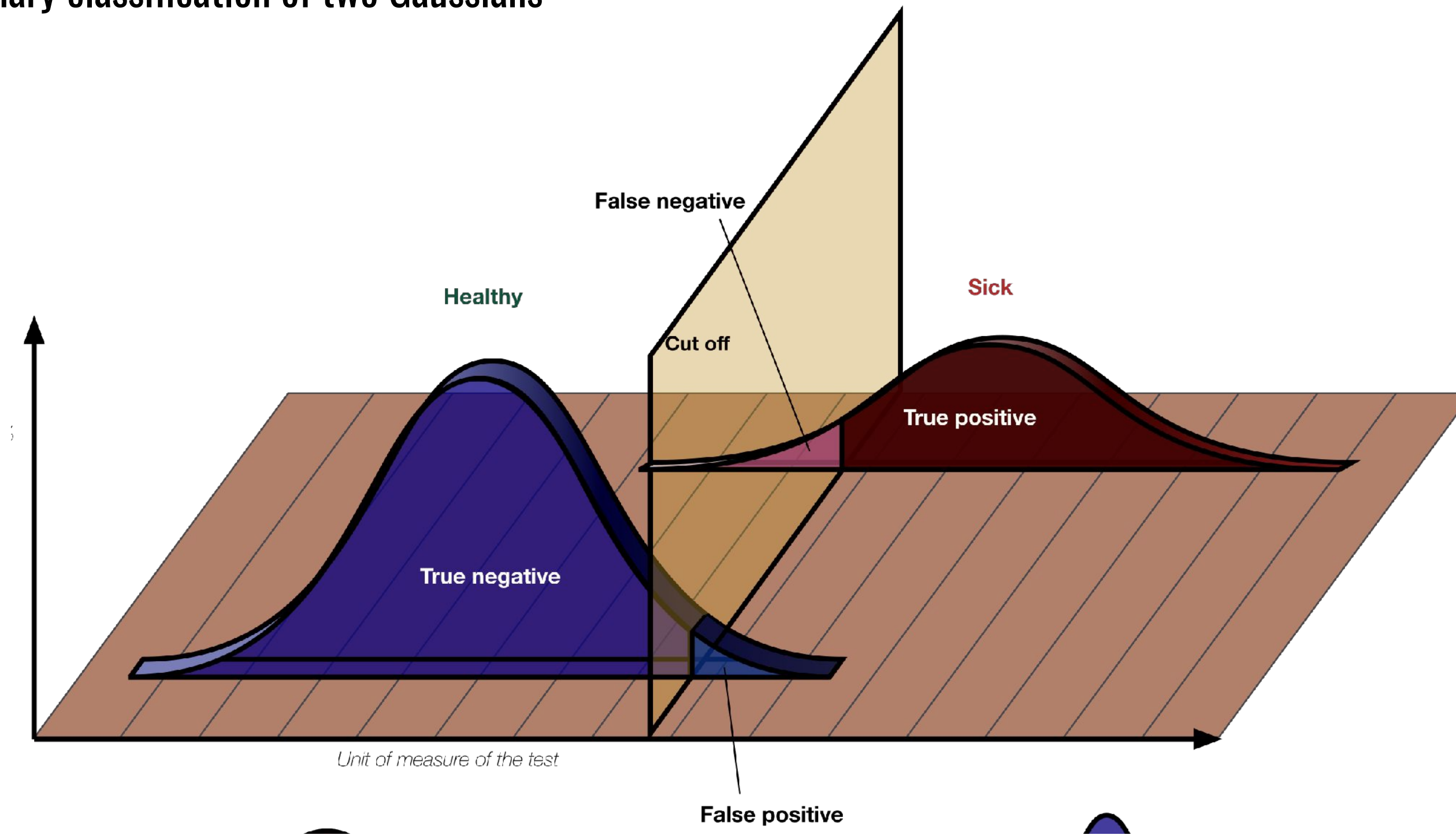
How many relevant items are retrieved?



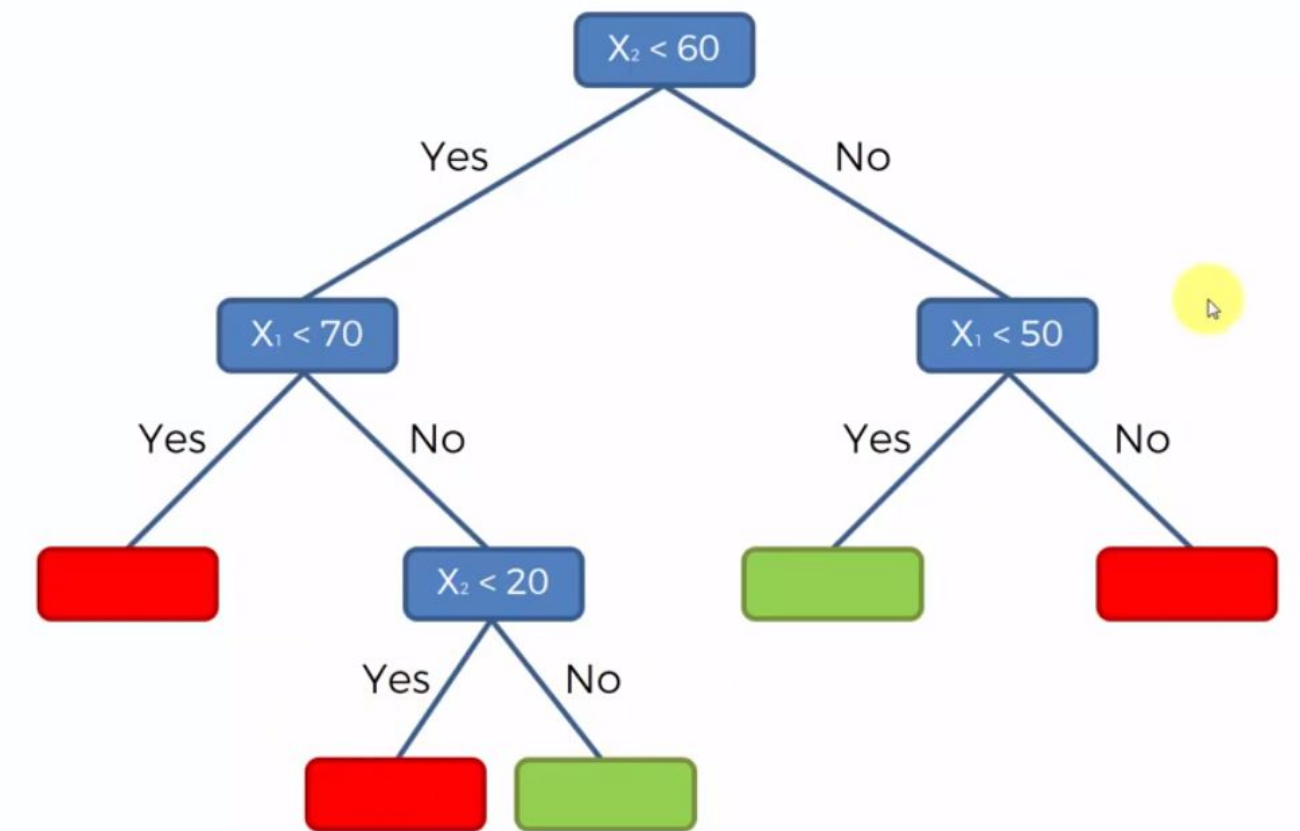
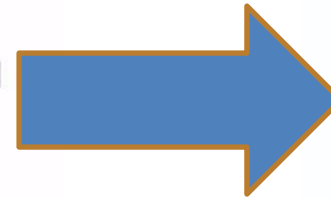
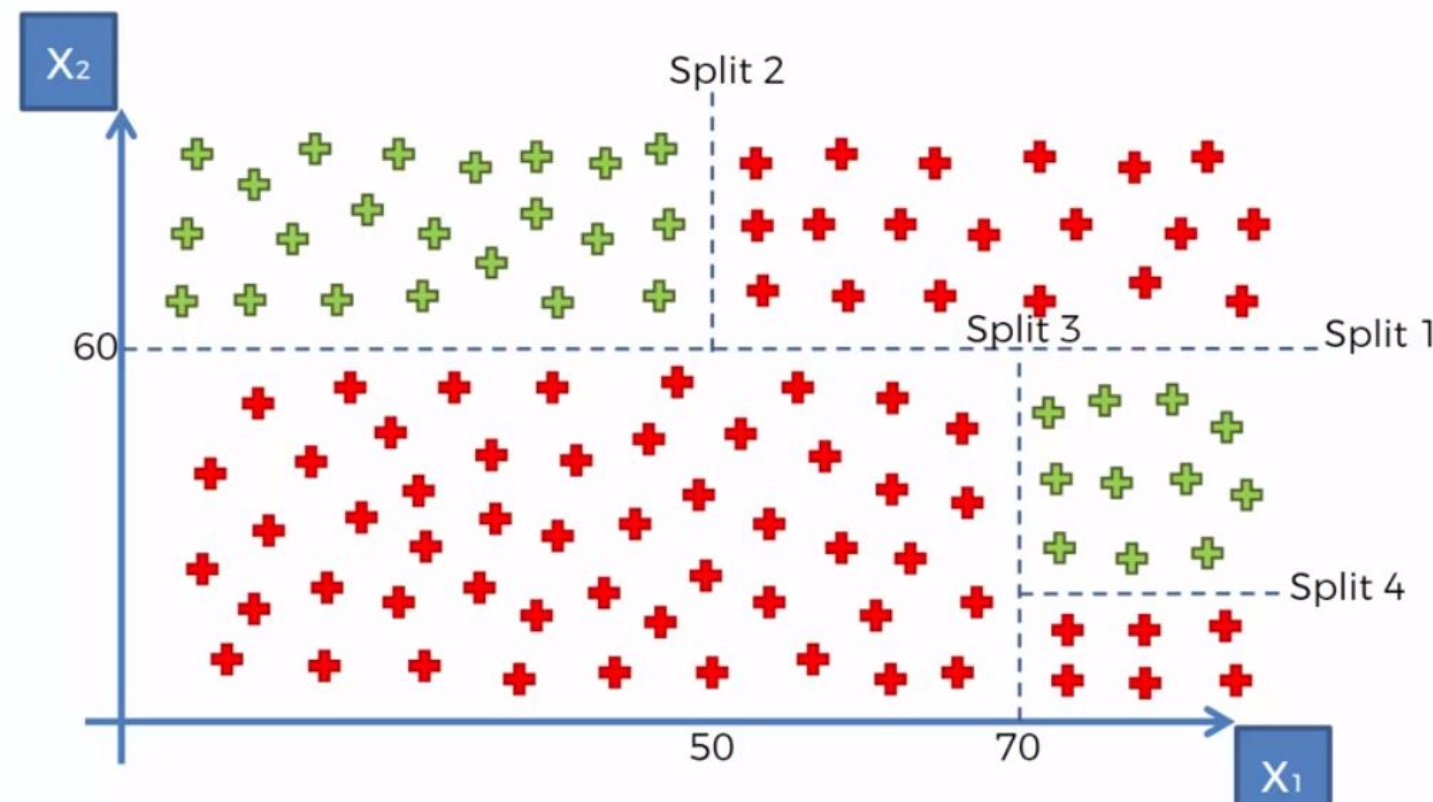
https://en.wikipedia.org/wiki/Precision_and_recall

Classification

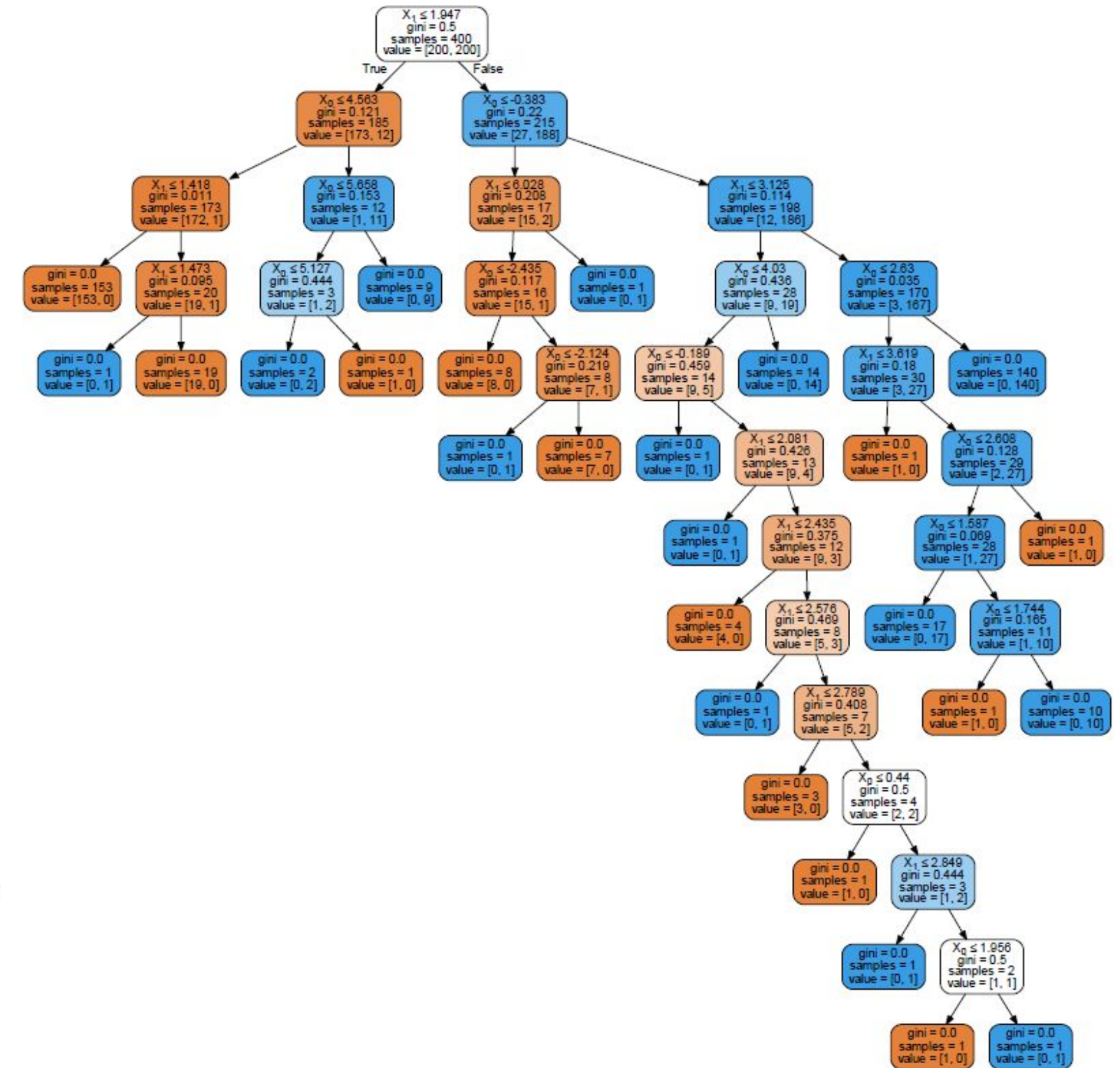
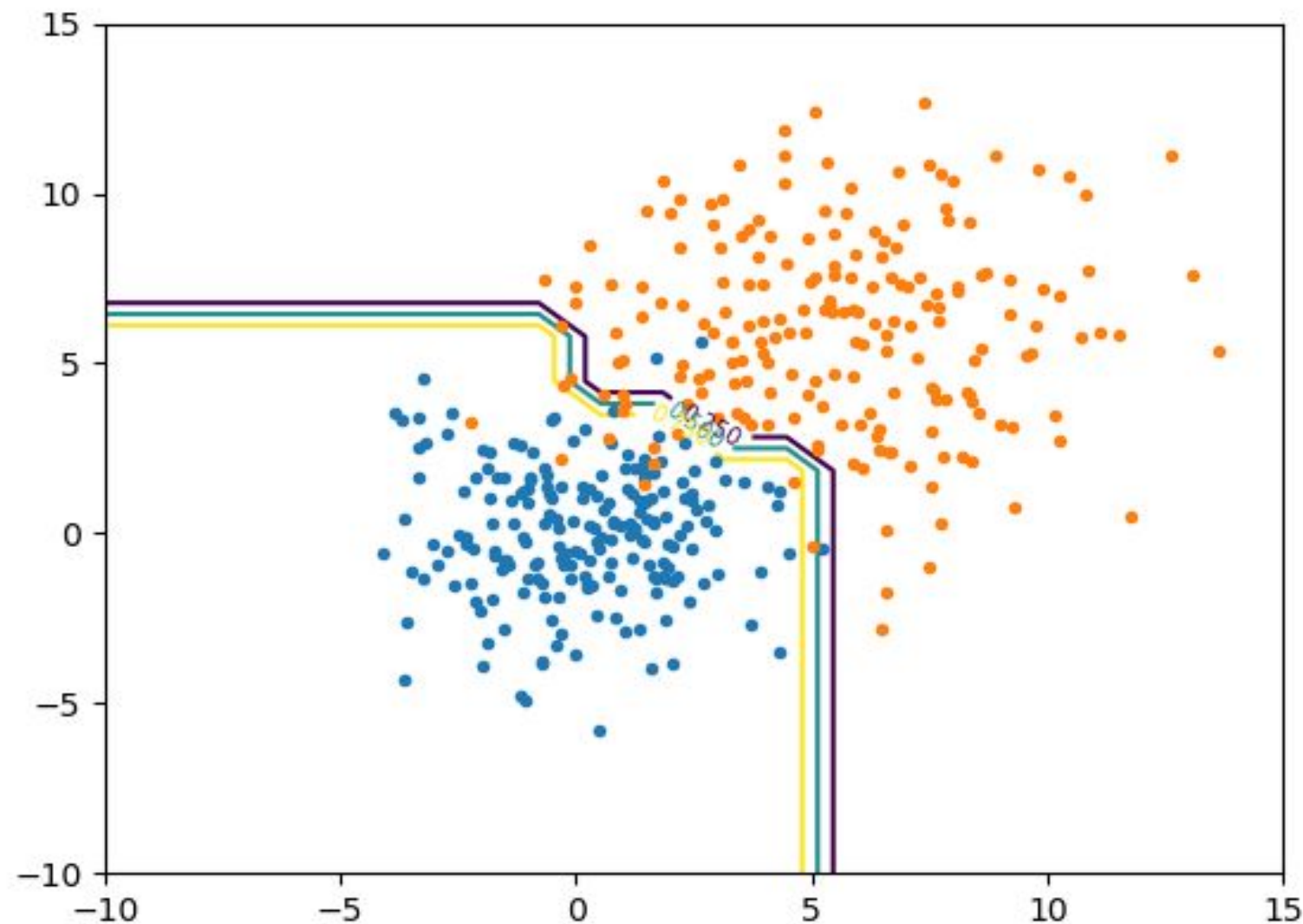
Binary classification of two Gaussians



Decision Trees

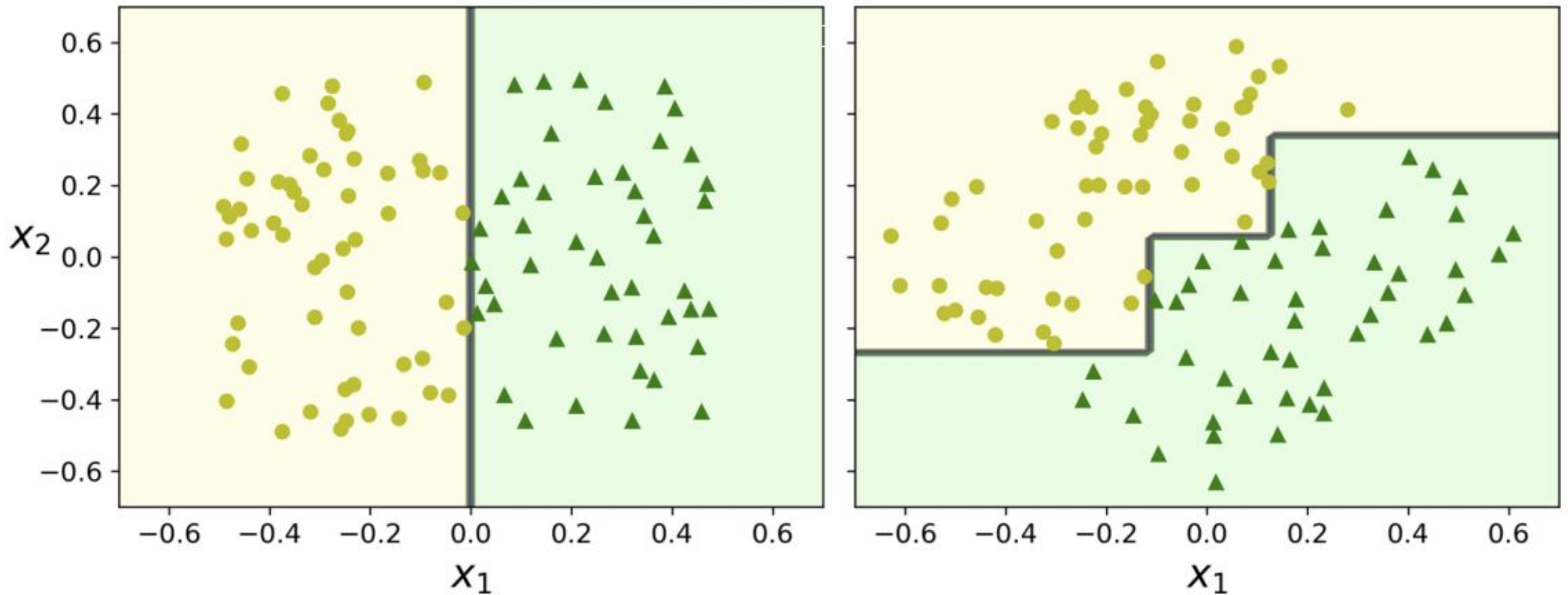


Supervised Learning: Decision Trees



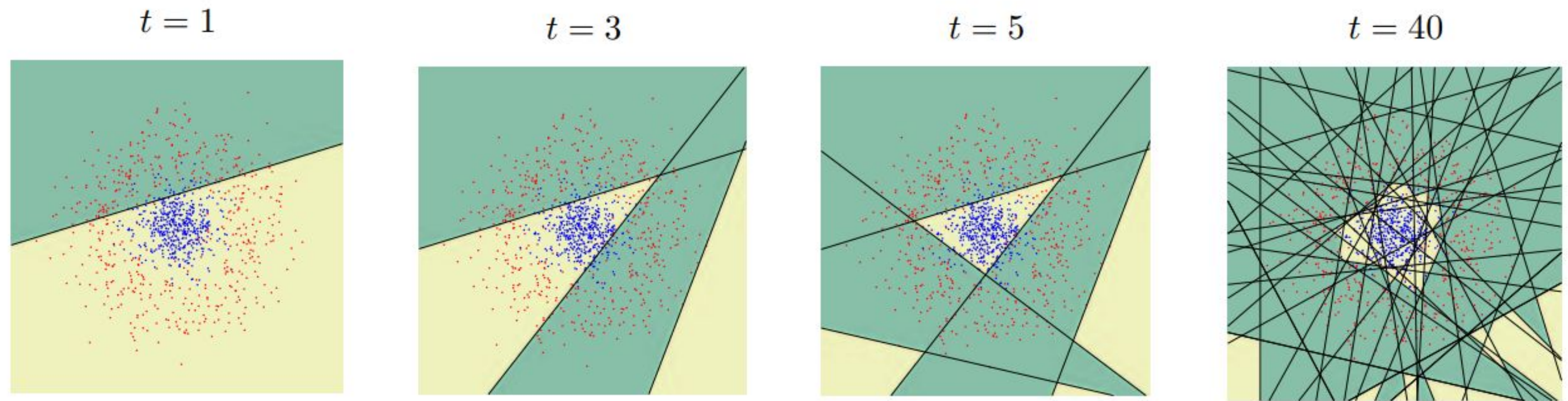
Supervised Learning: Decision Trees

Sensitivity to data representation



Ensemble Learning

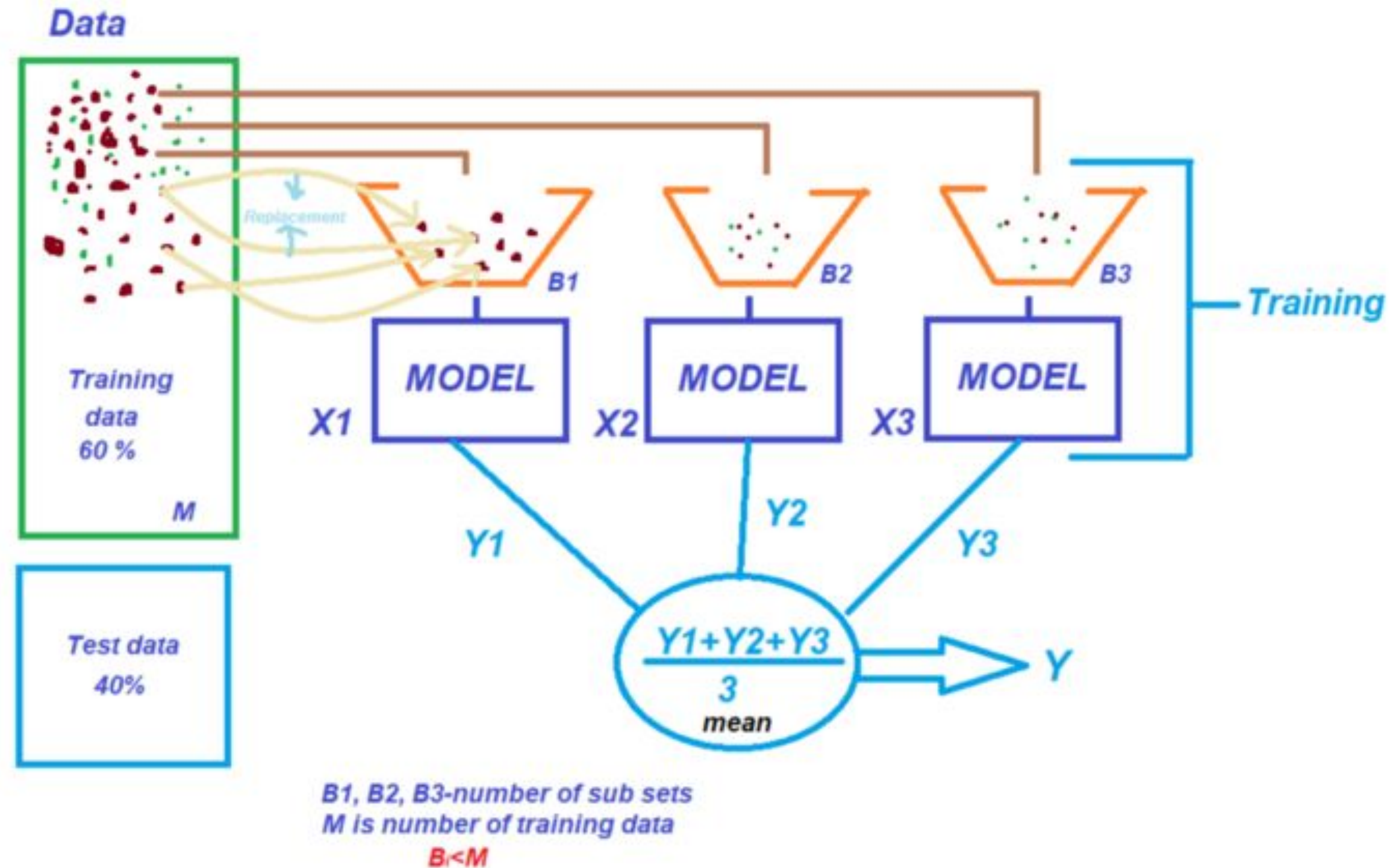
Boosting



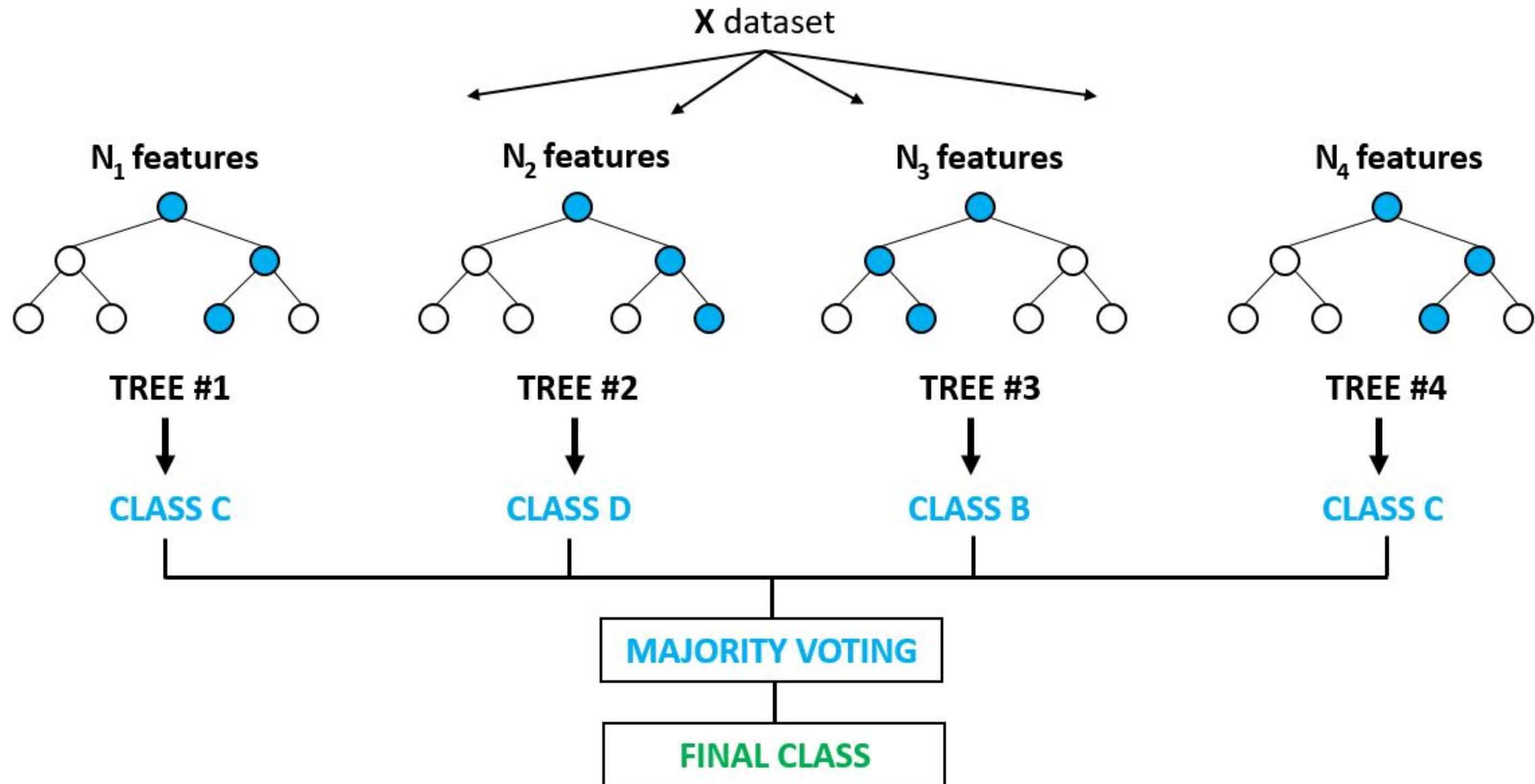
AdaBoost

Ensemble Learning

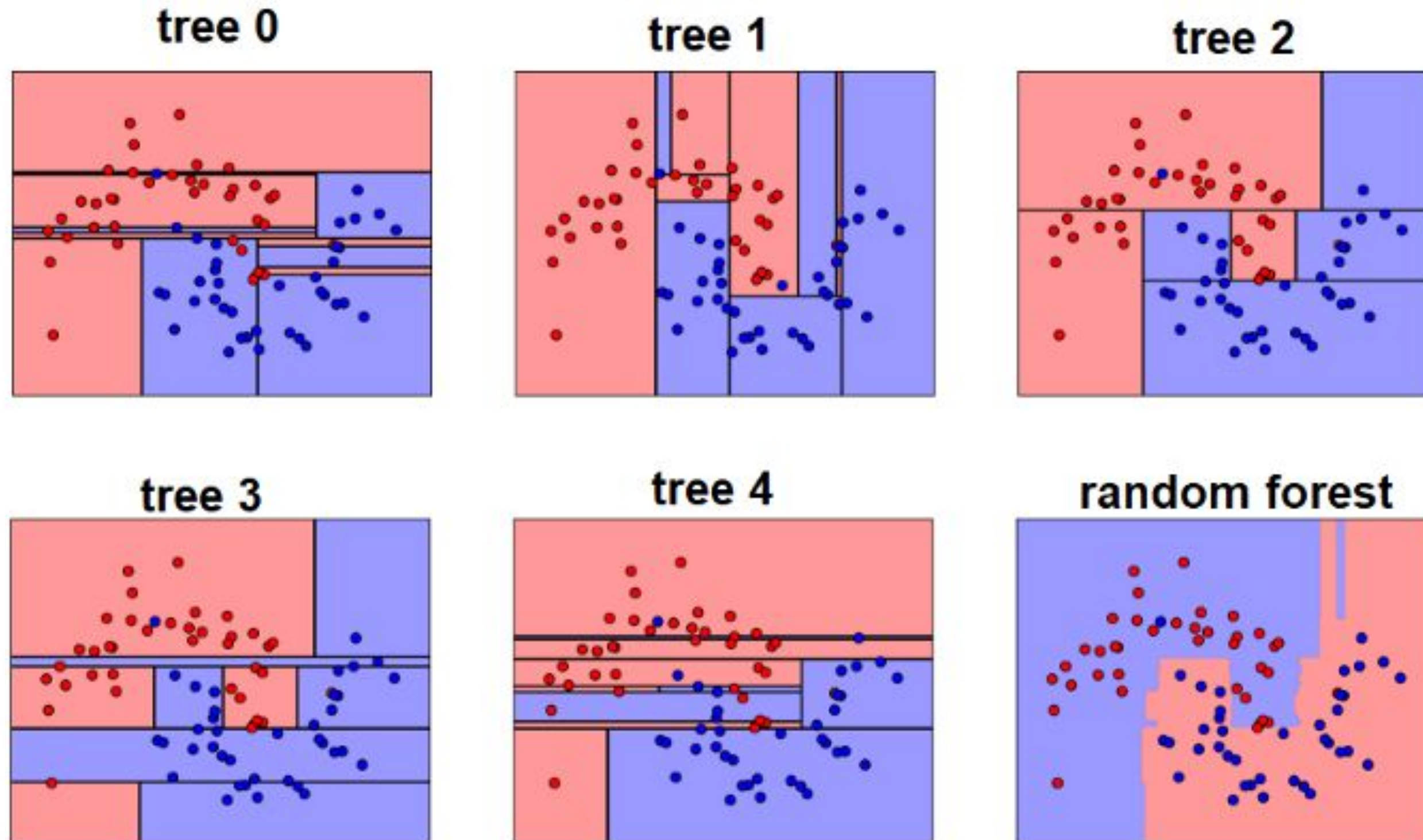
Bagging (bootstrap)



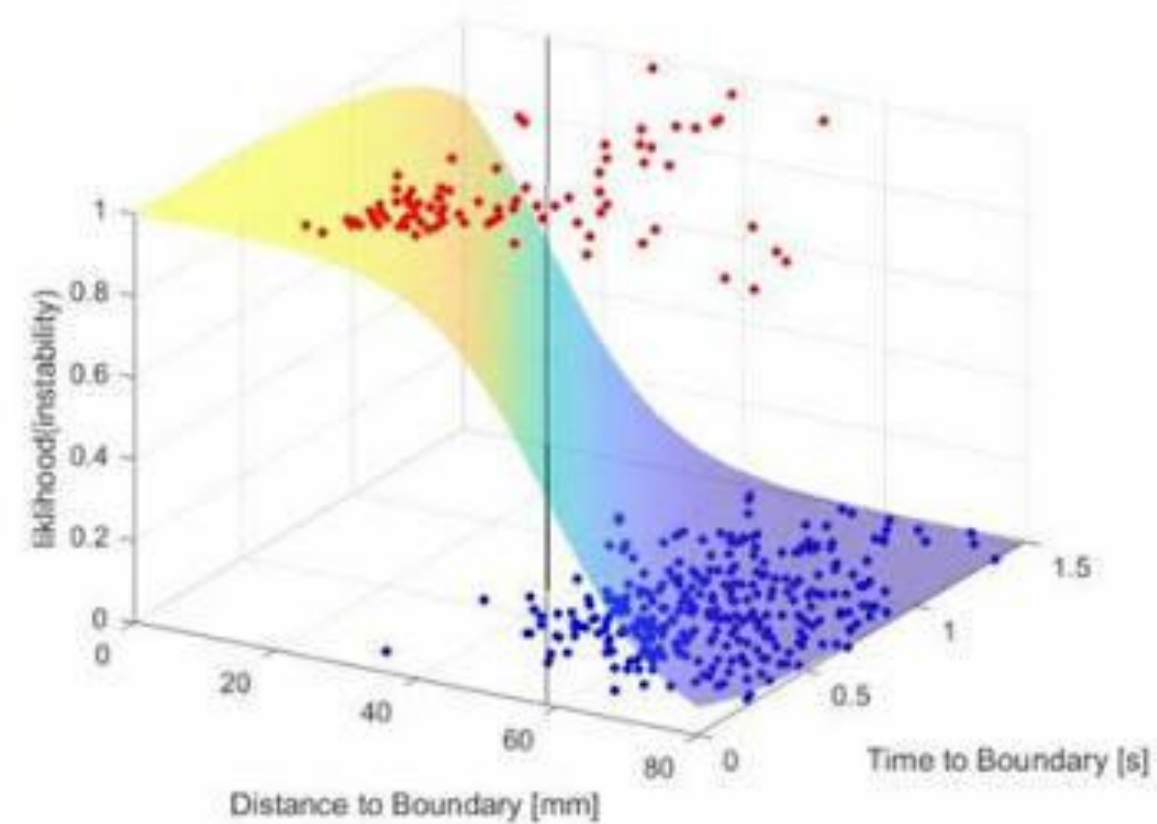
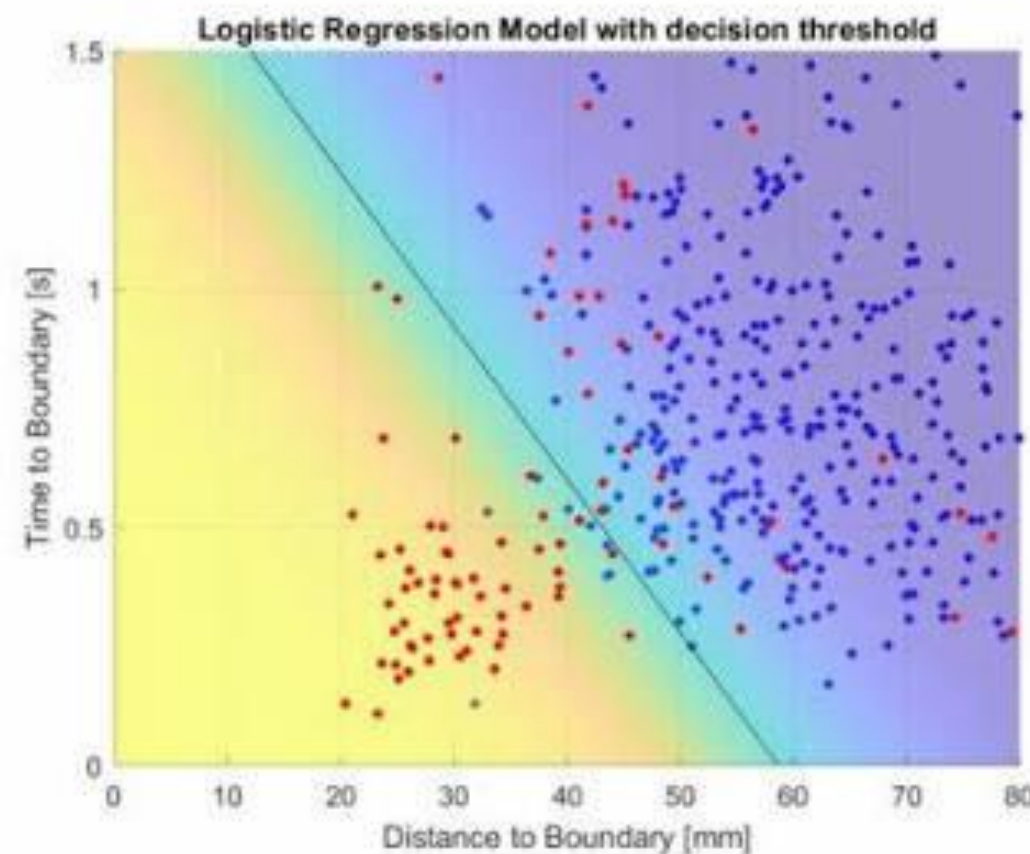
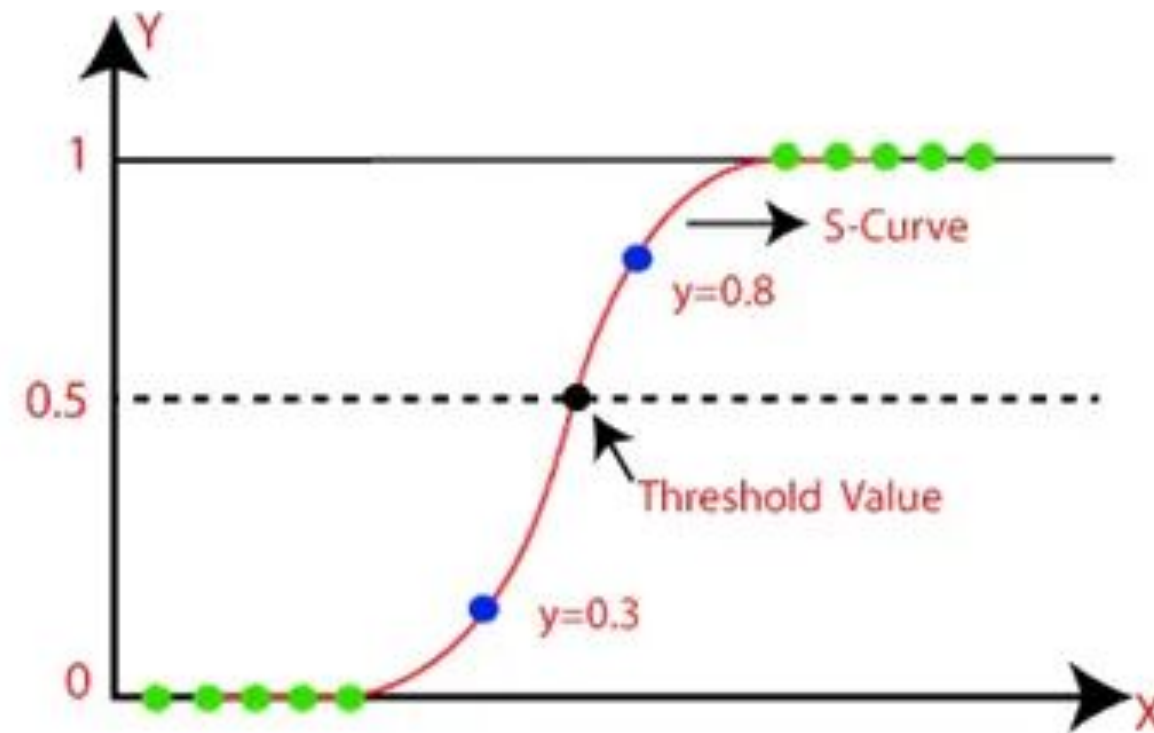
Random Forest



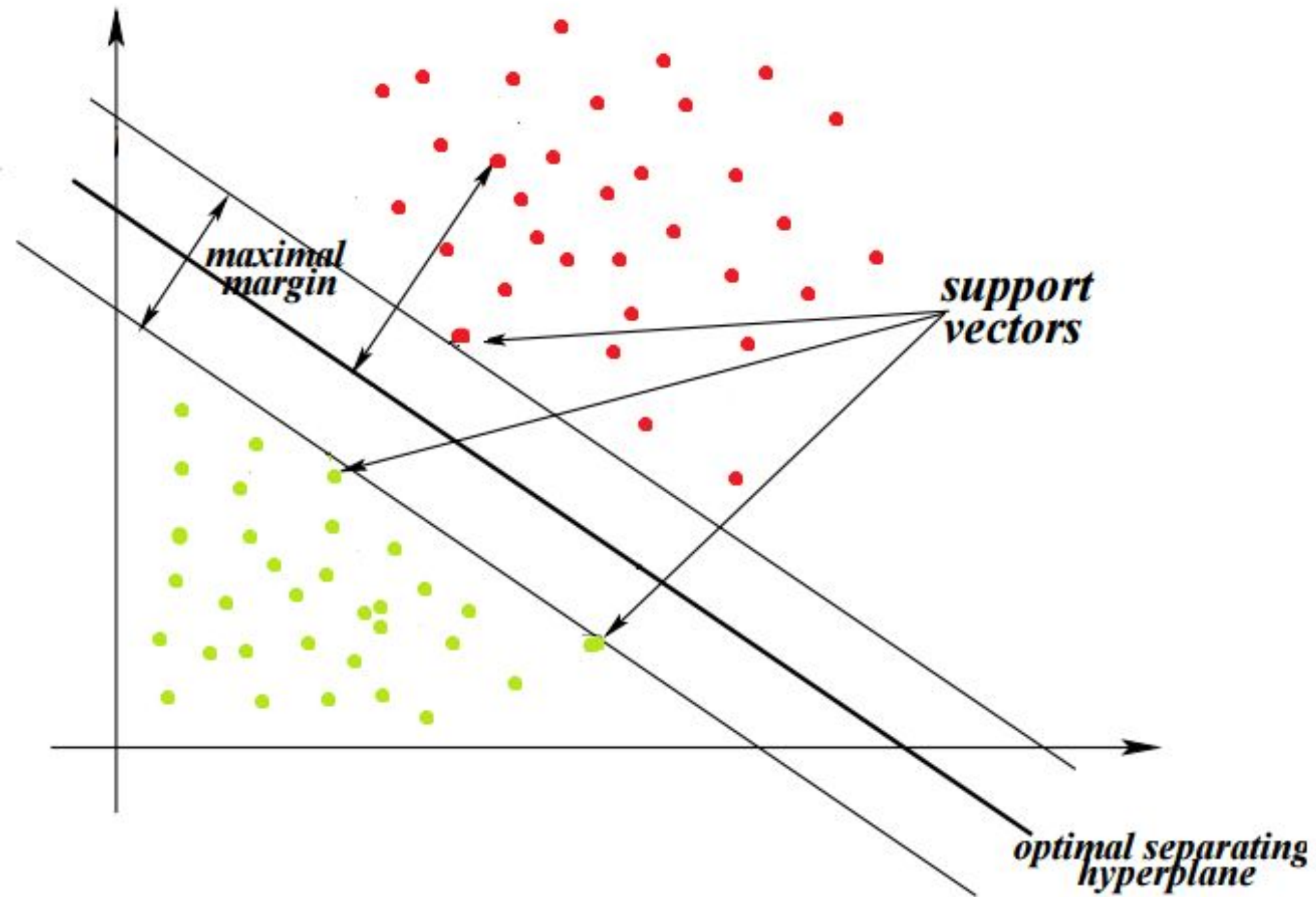
Random Forest



Logistic Regression

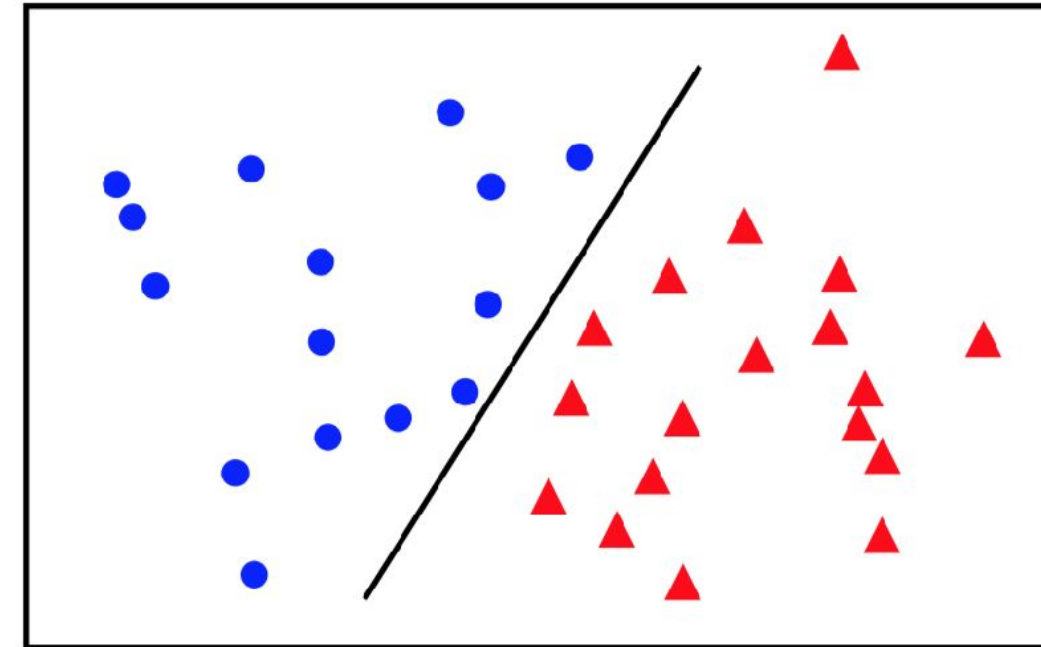
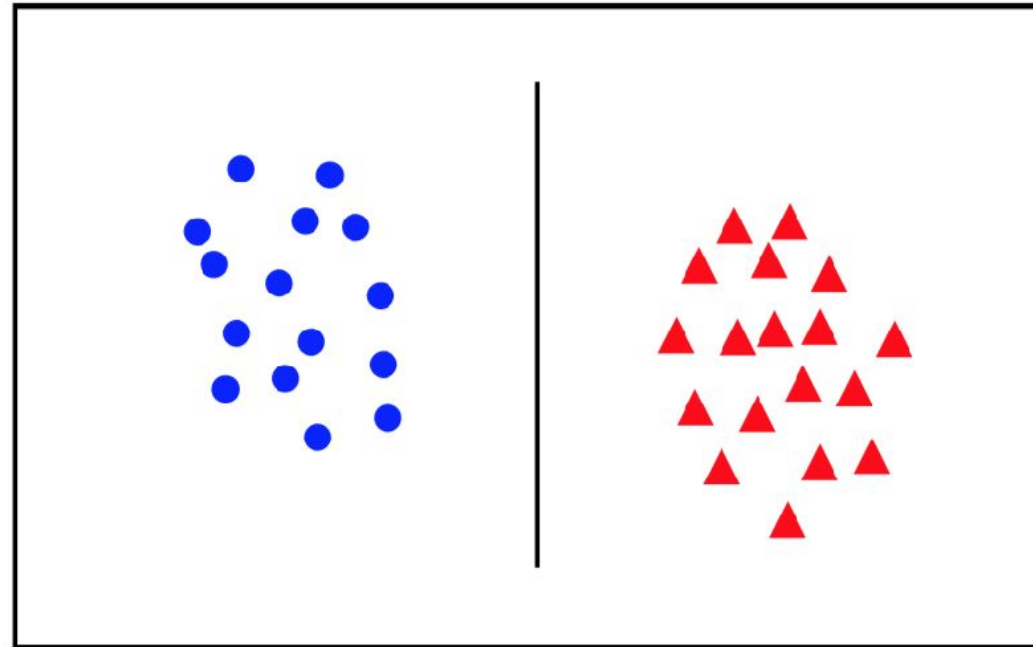


SVM = Support Vector Machine

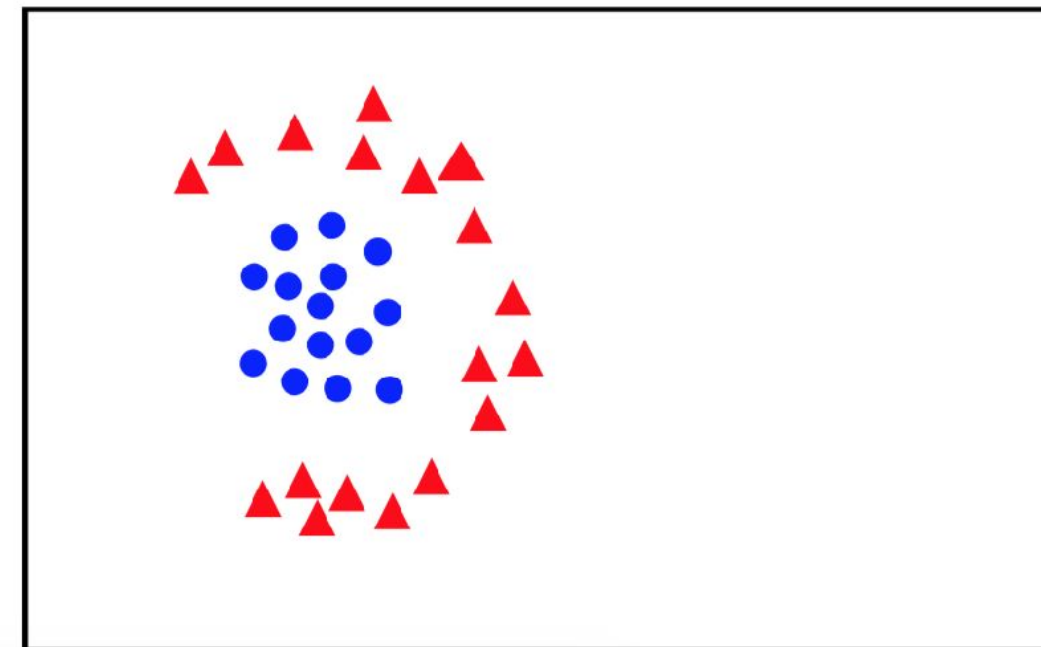
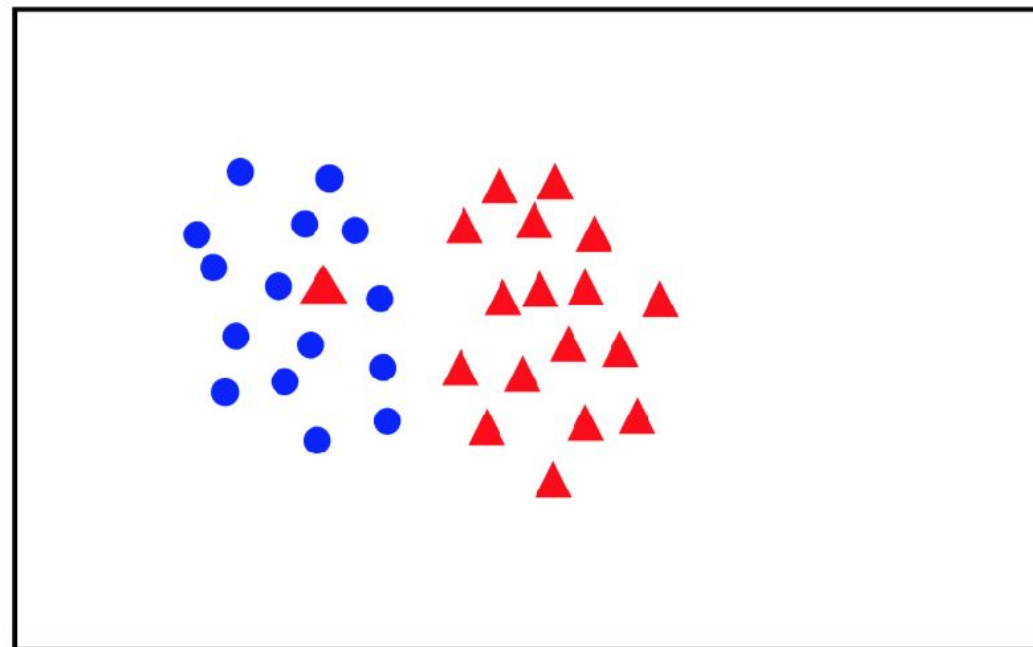


Separability of Data

linearly
separable



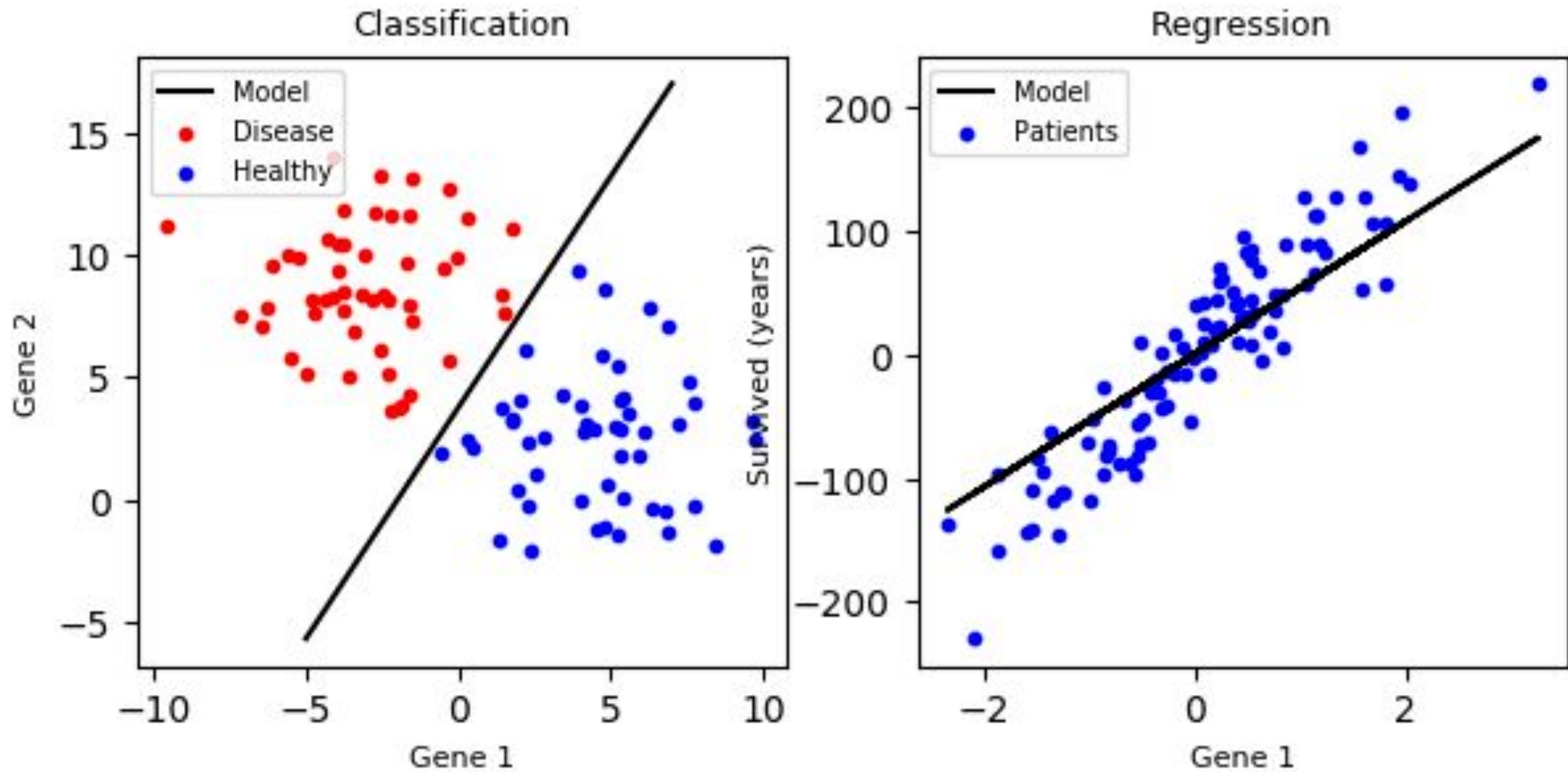
not
linearly
separable



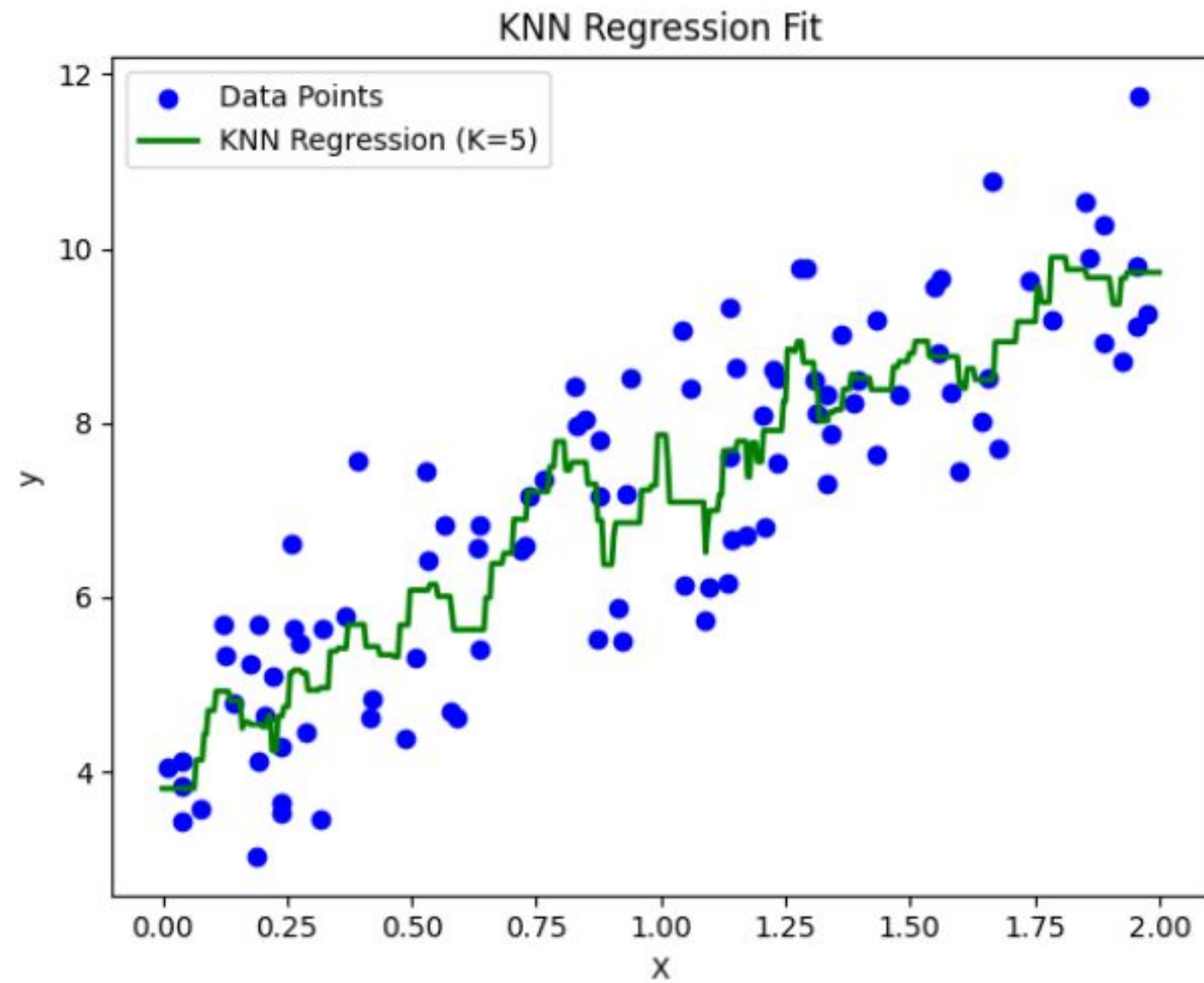
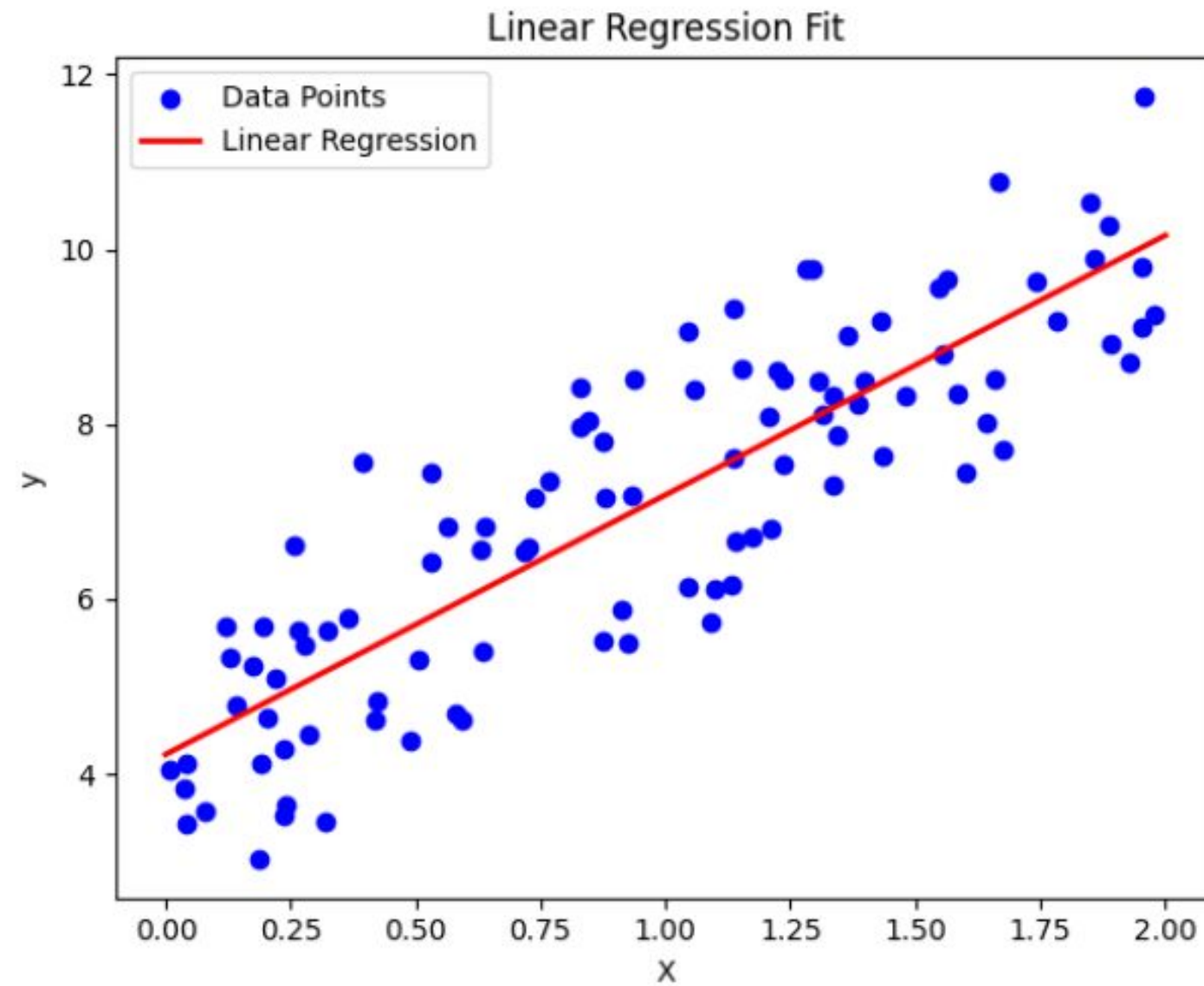
Model types

- **Linear models** $y = a_i * x_i + b$
- **Nonlinear models** $y = a_i * x_i^n + b_i * x_i + c$
 $y = a$ if $x > b$
- **Not a model at all**

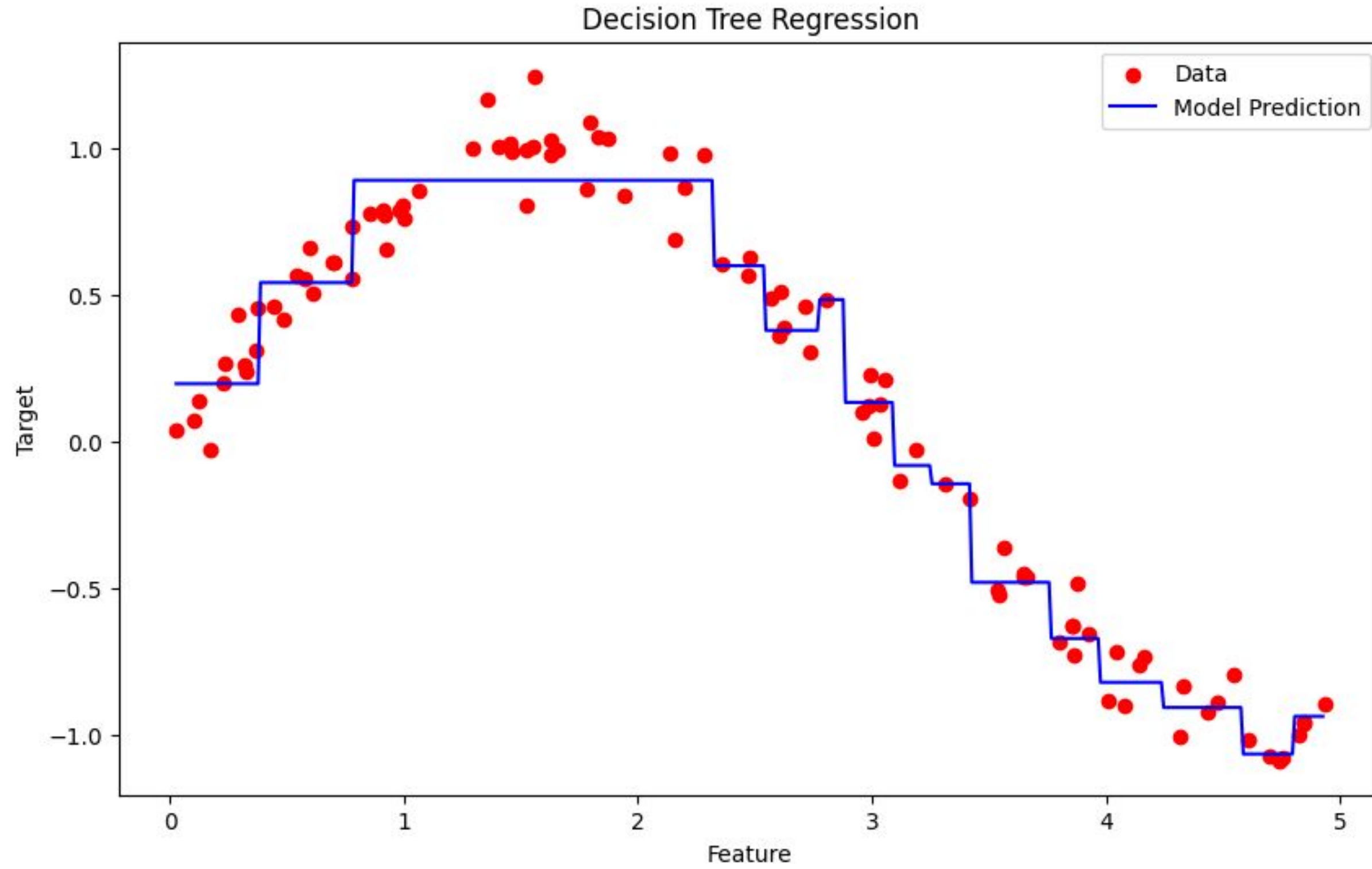
Regression



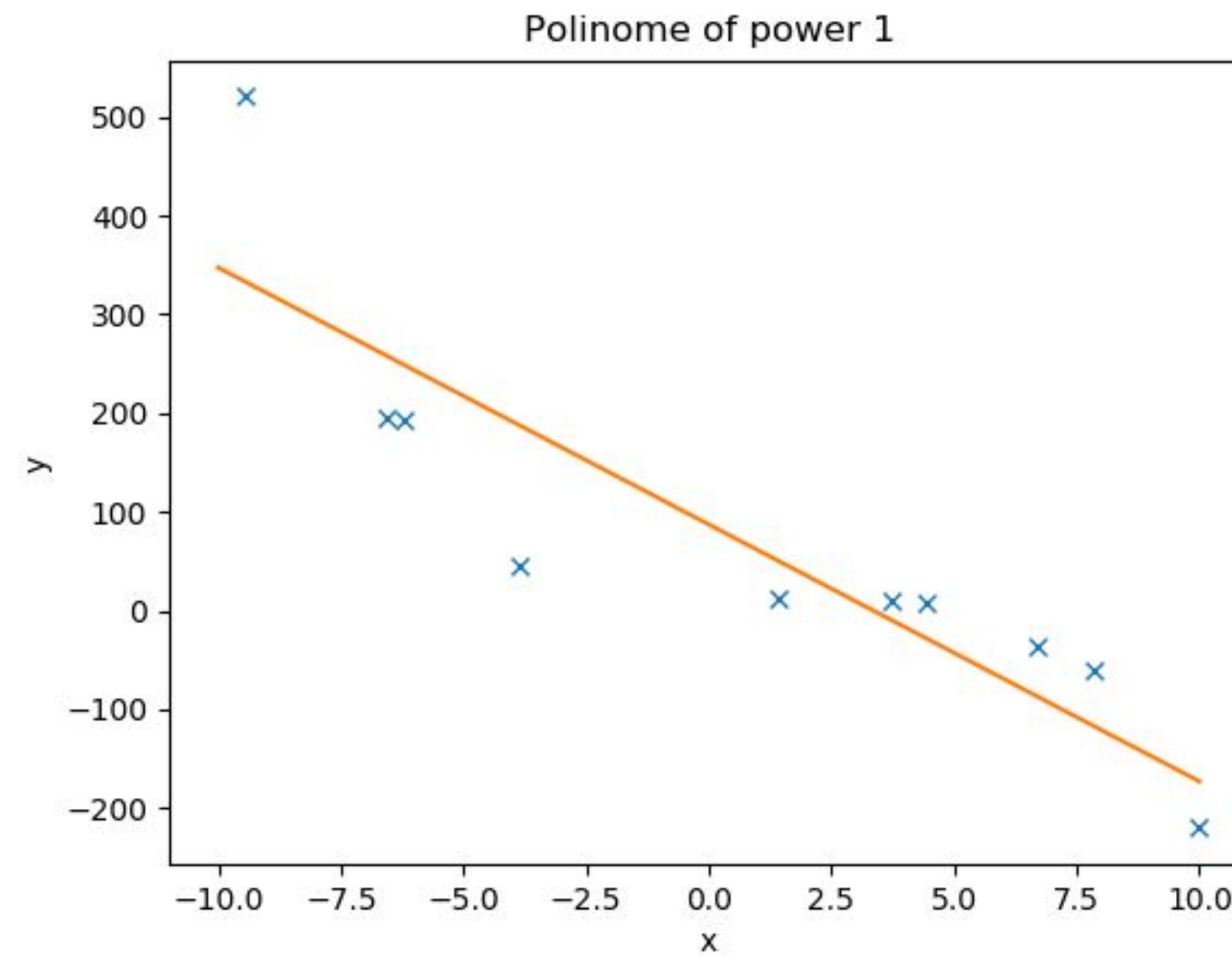
Regression



Regression



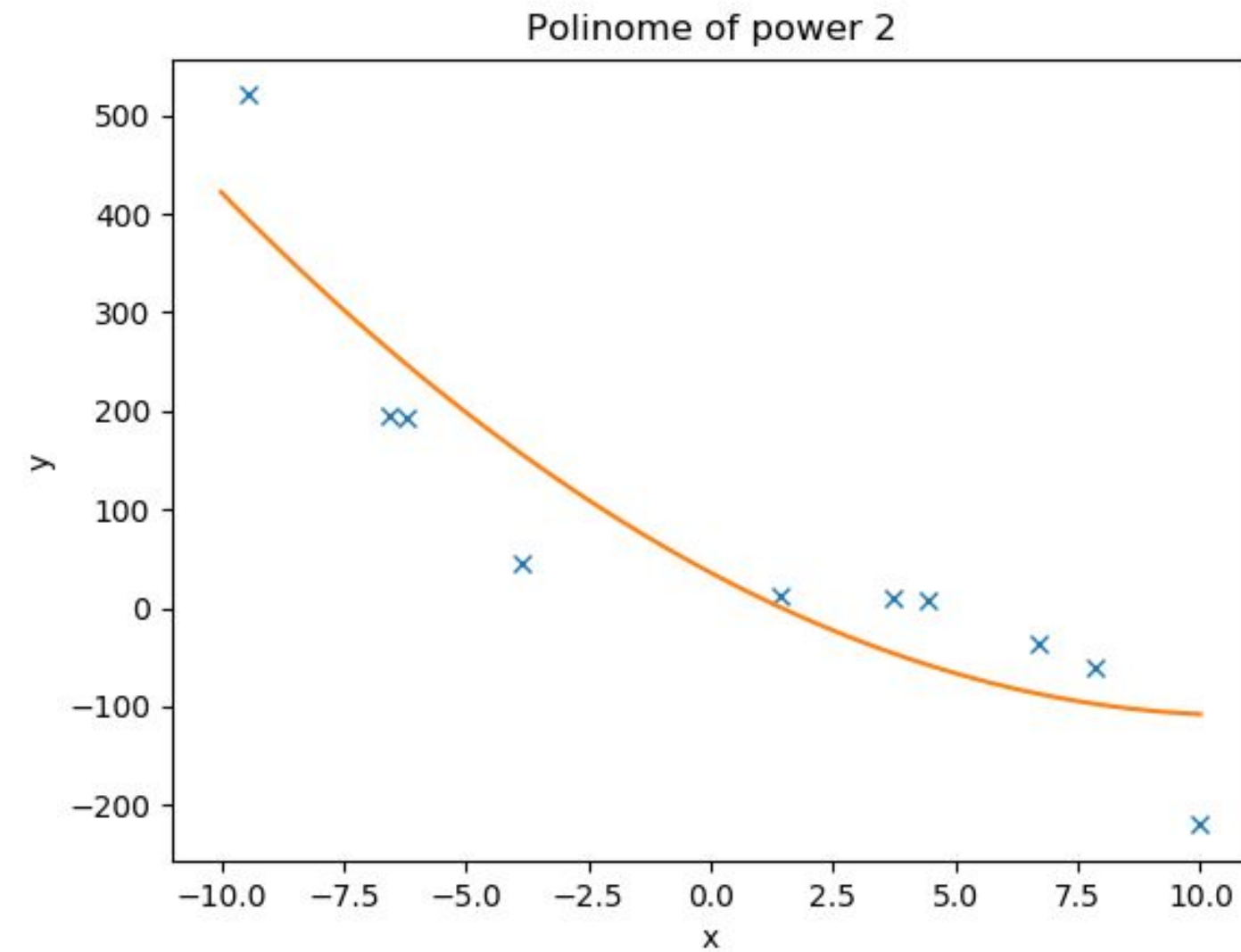
Regression

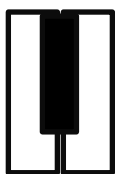


a  b

Okay, not bad

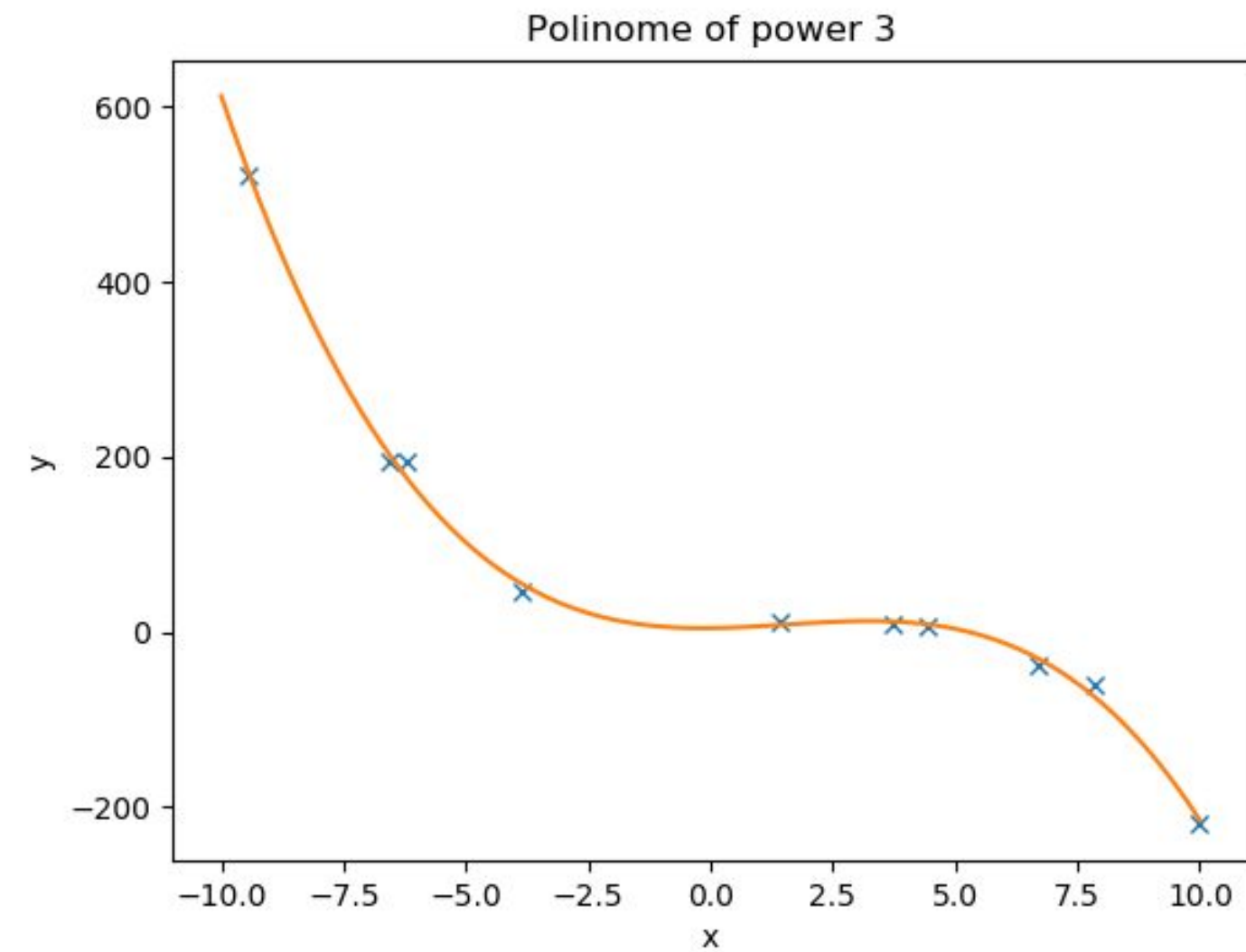
Regression

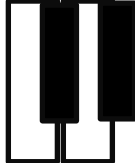


a 
b c

Better!

Regression

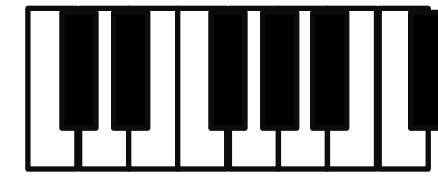
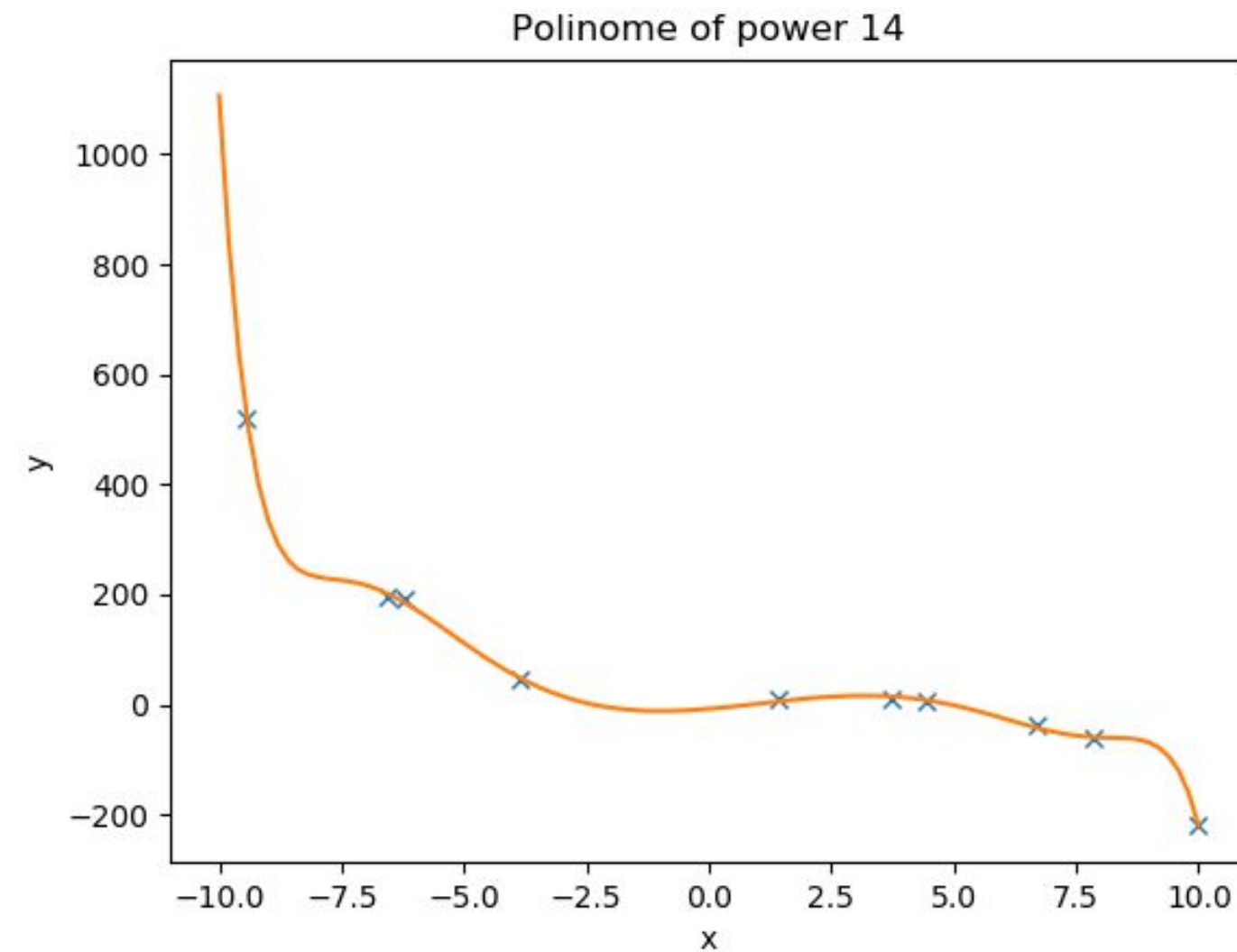


a 
b c d

Woow! Let's add some more!



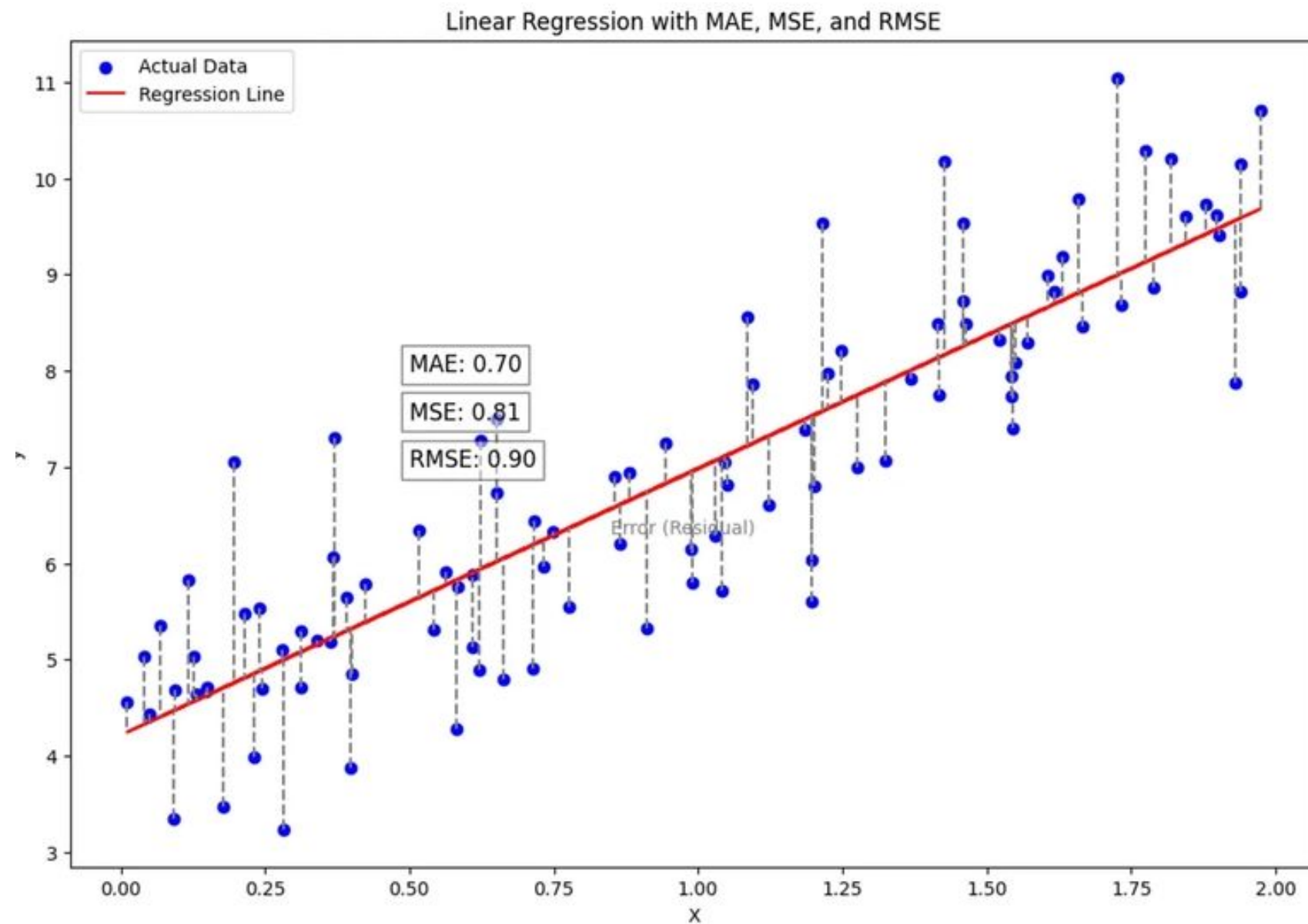
Regression



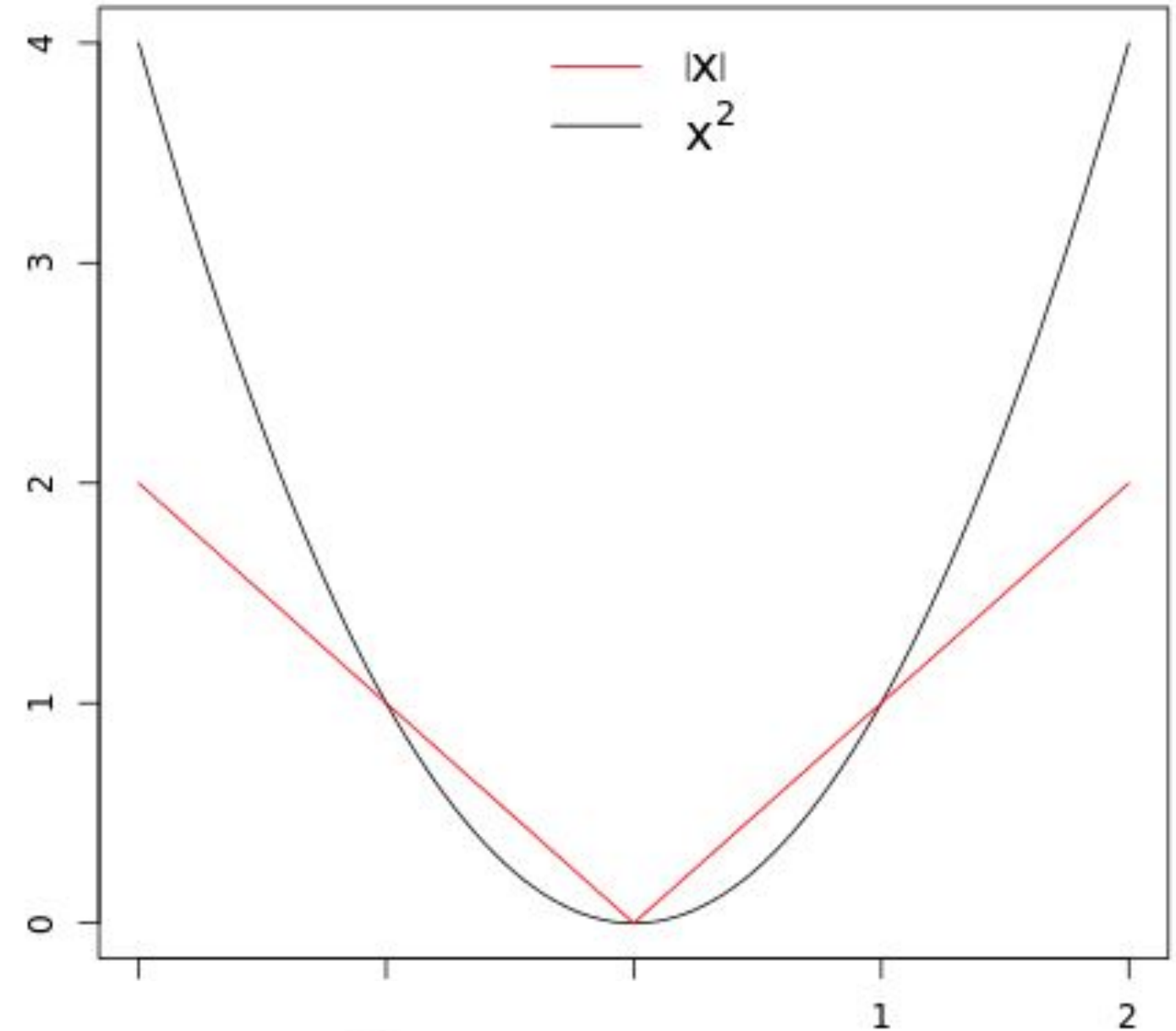
Maybe it is a good approximation. But how can we be sure?

Regression

Regression main metrics

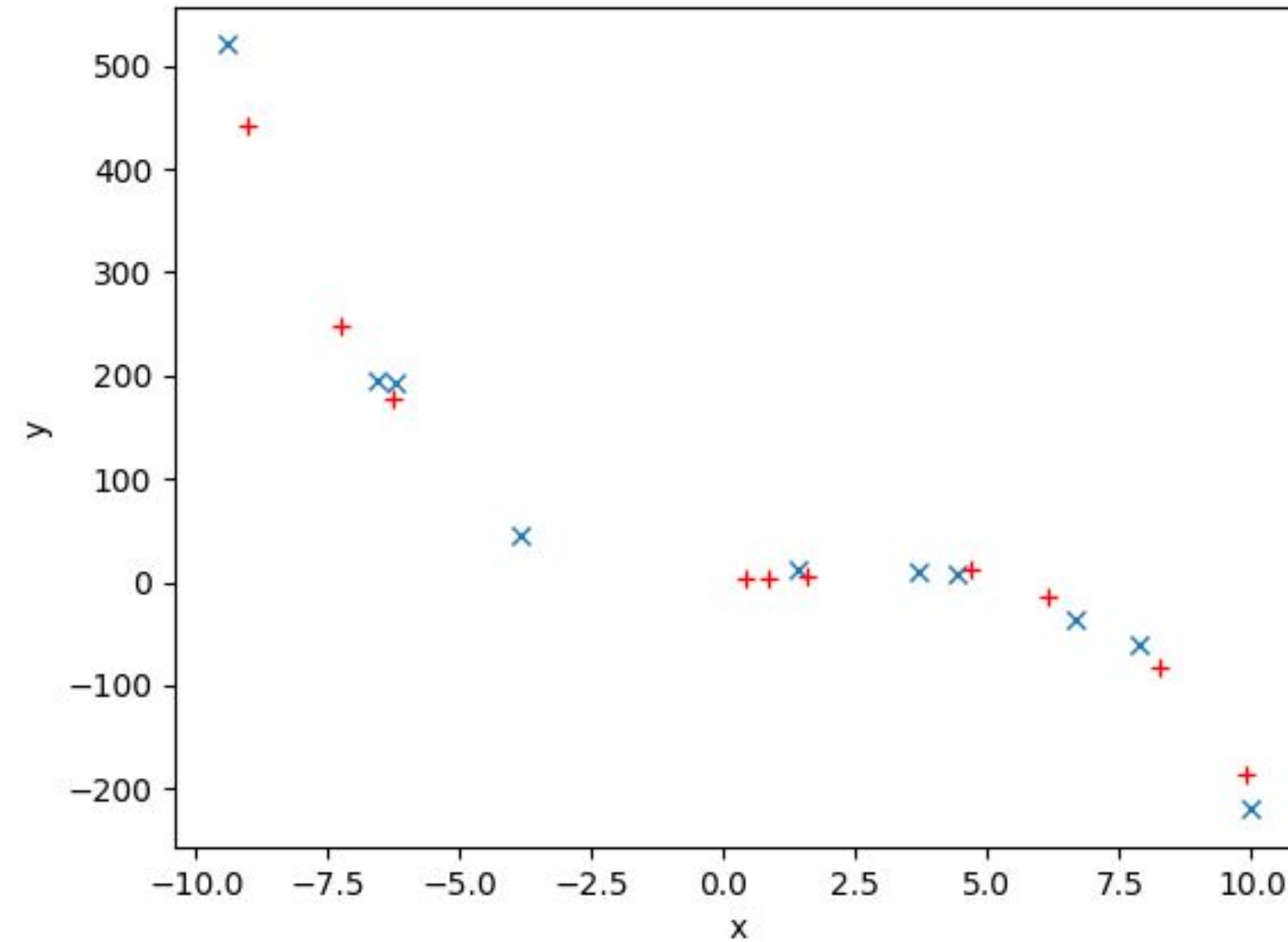


$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}|$$



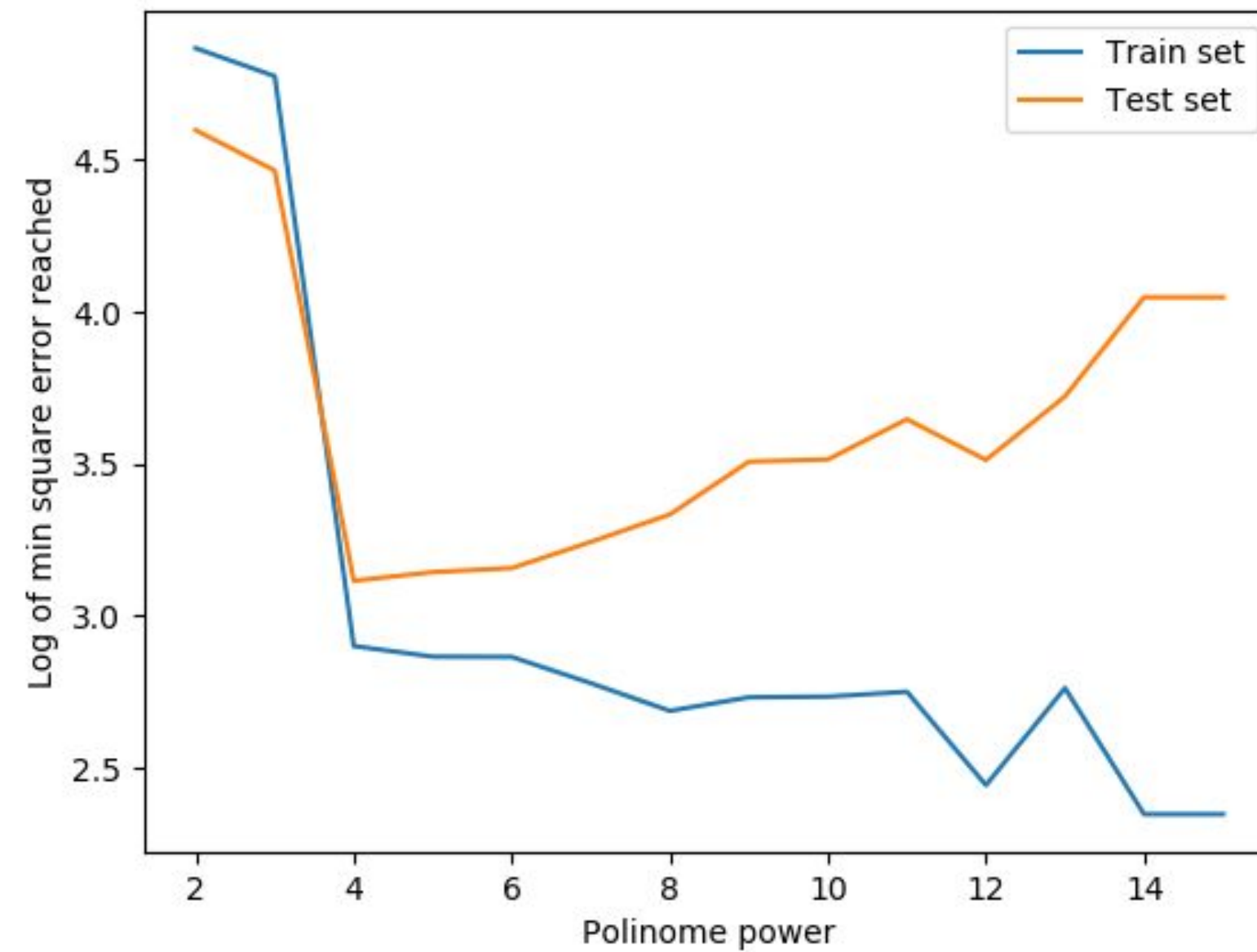
$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2$$

Regression



I will ask my customer to give me more data!

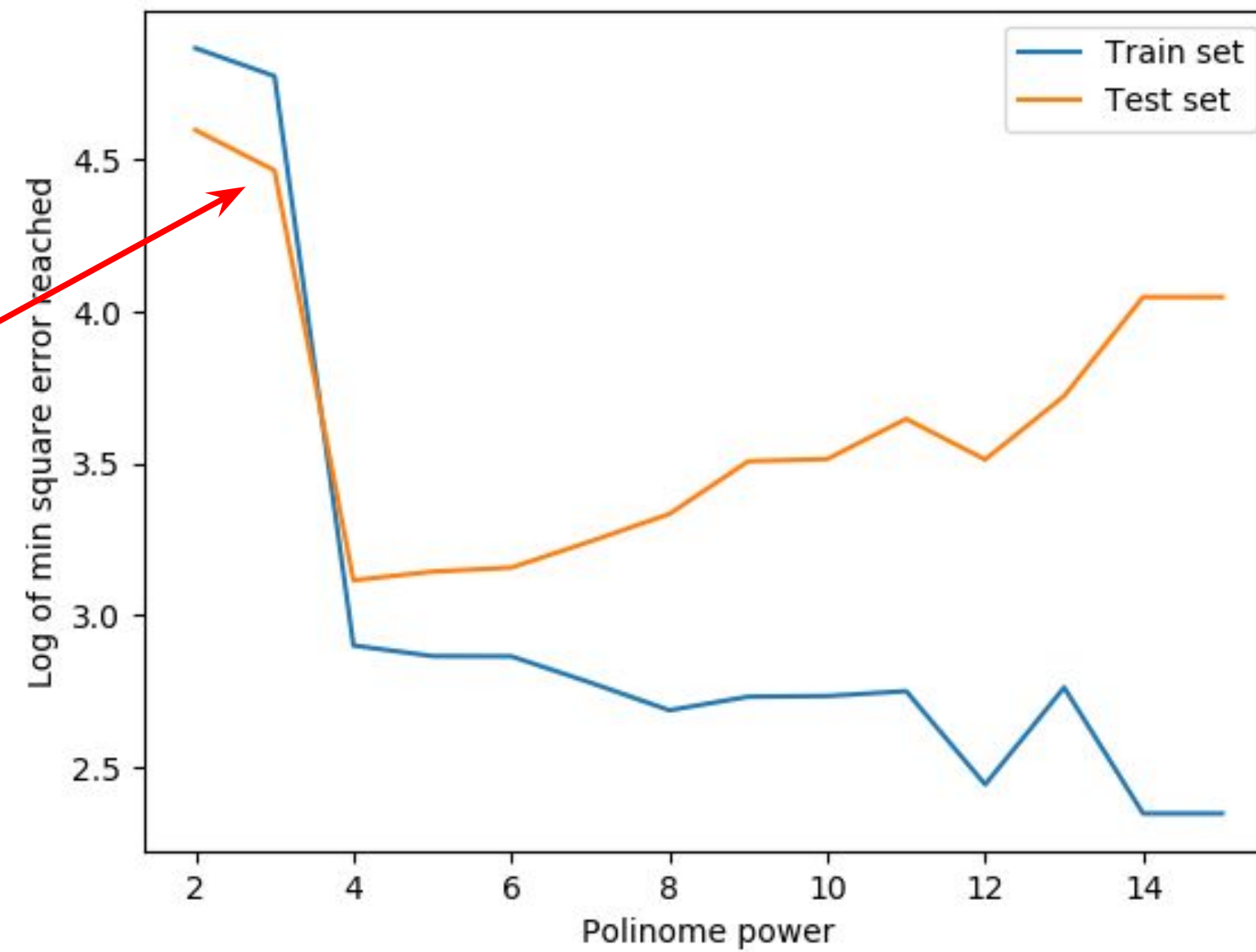
Regression



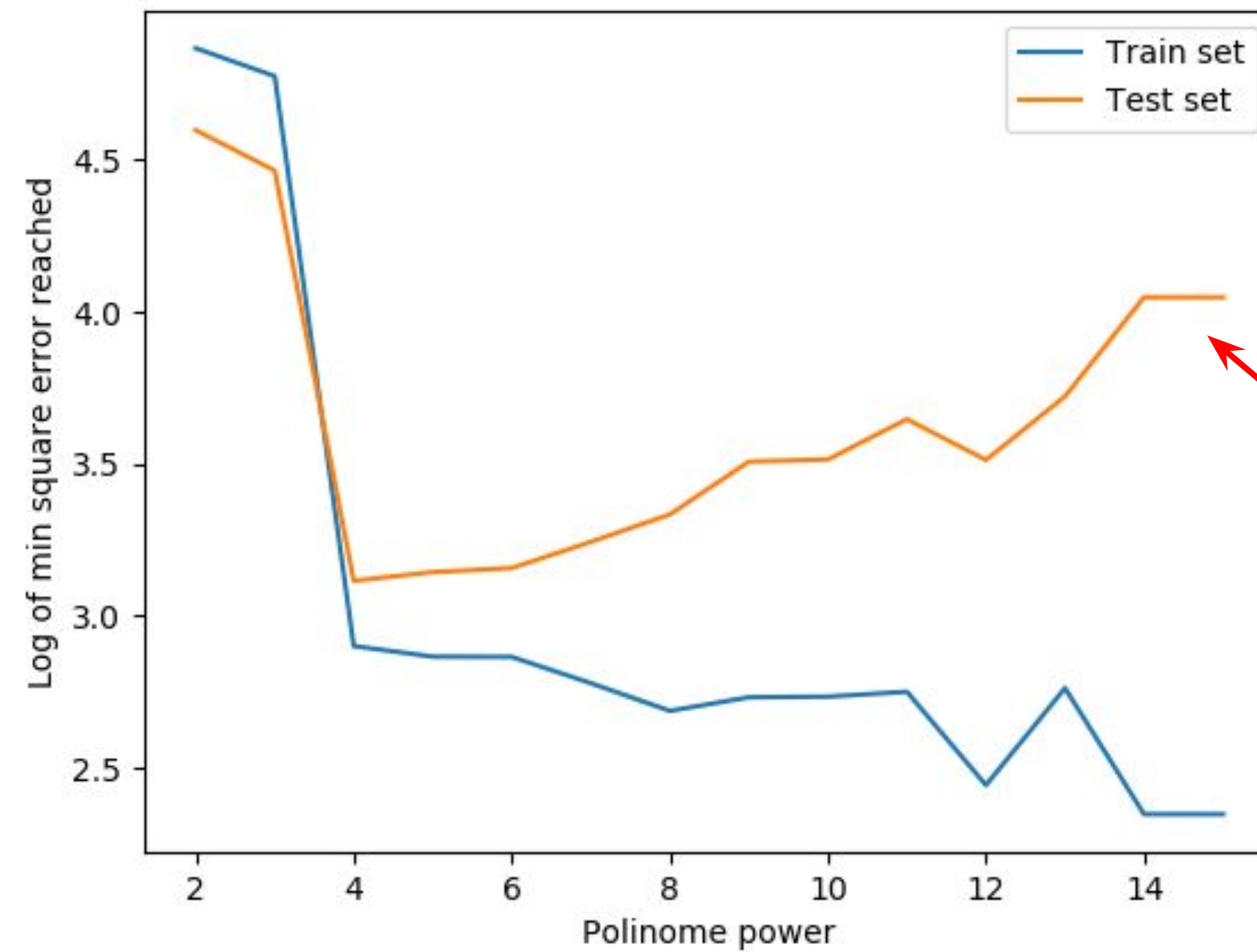
The model is learning better, but the result is worse!

Regression

Too simple model.
Won't learn
complexity of
our world



Regression



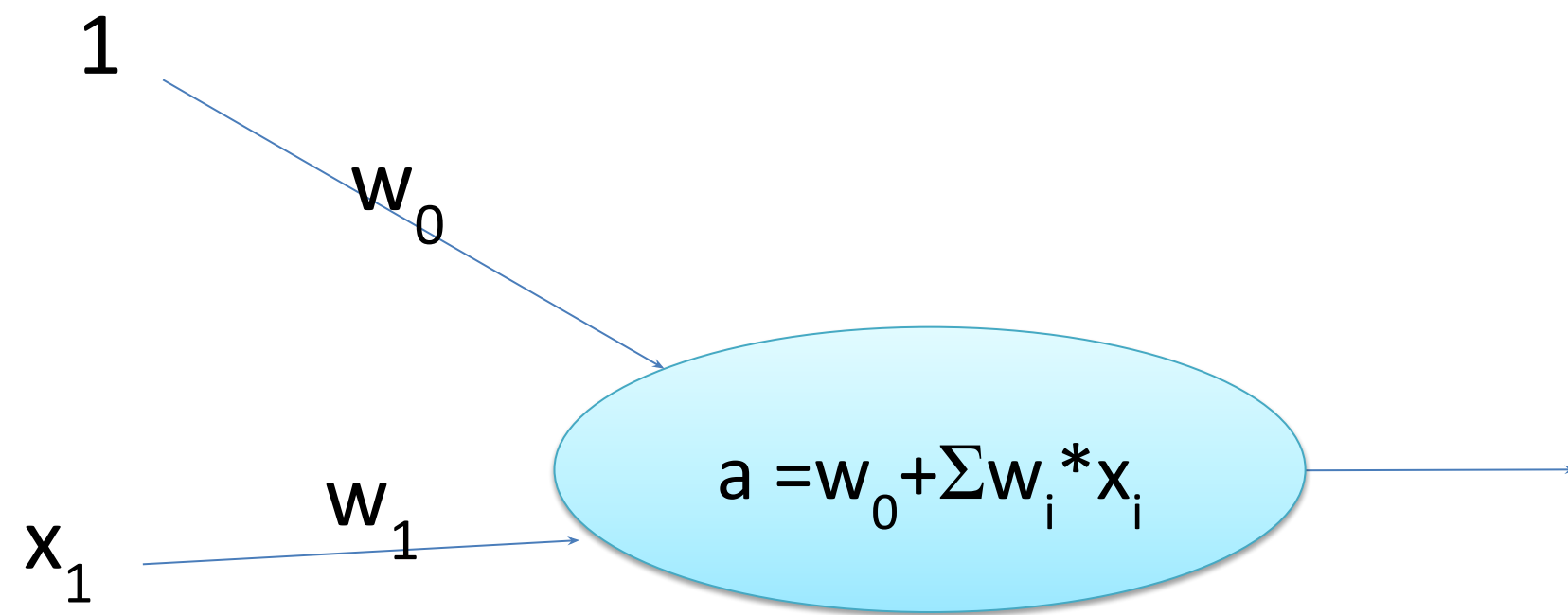
Too detailed model.
Learned our data instead
of the world.
OVERFIT

Dataset Preparation

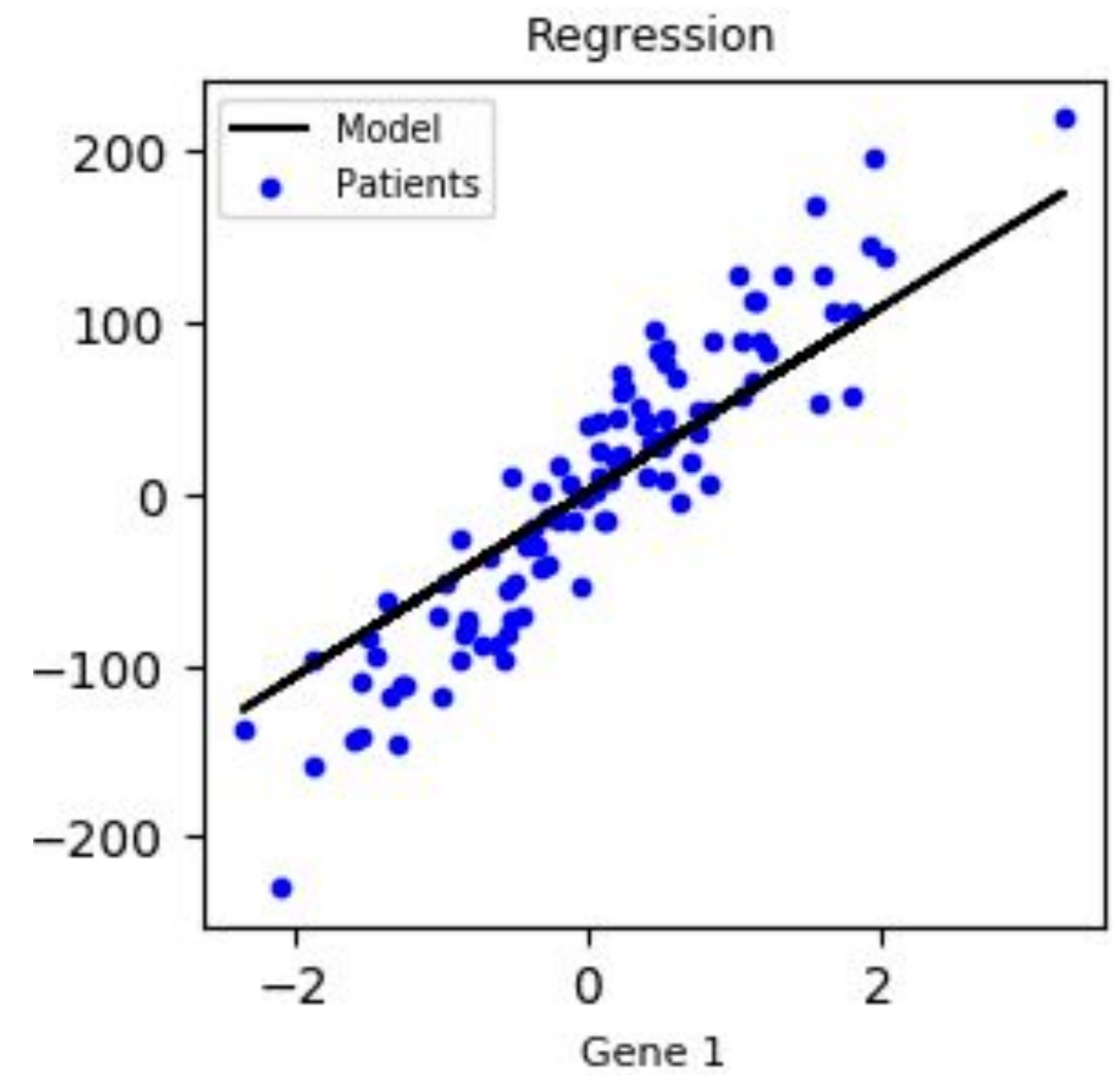
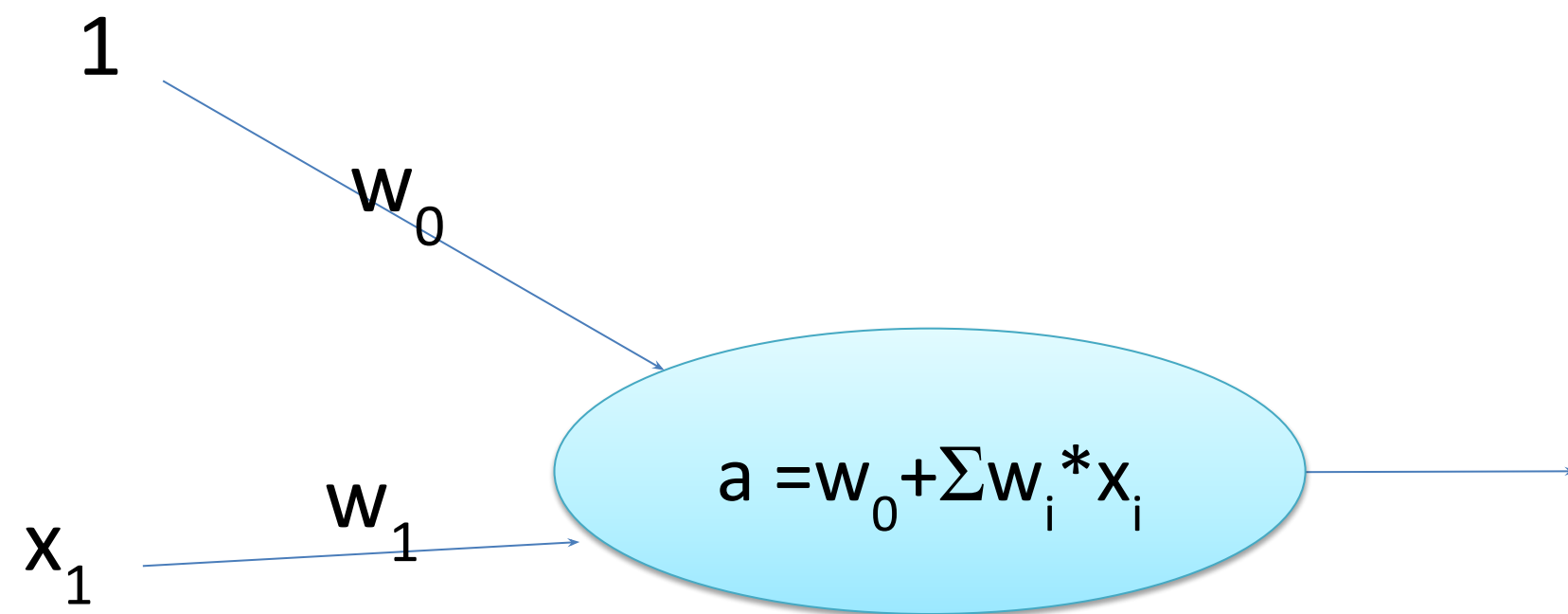
- Dataset organization: Training, Test, Validation



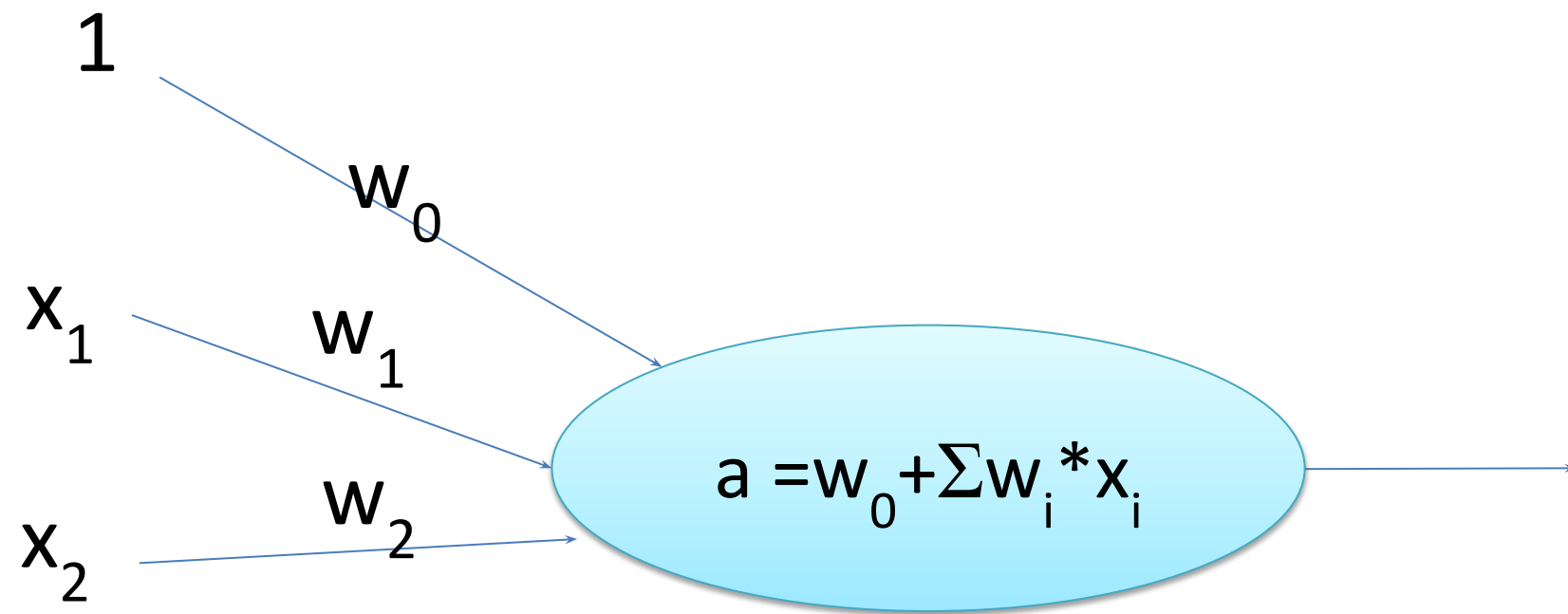
Neural Networks



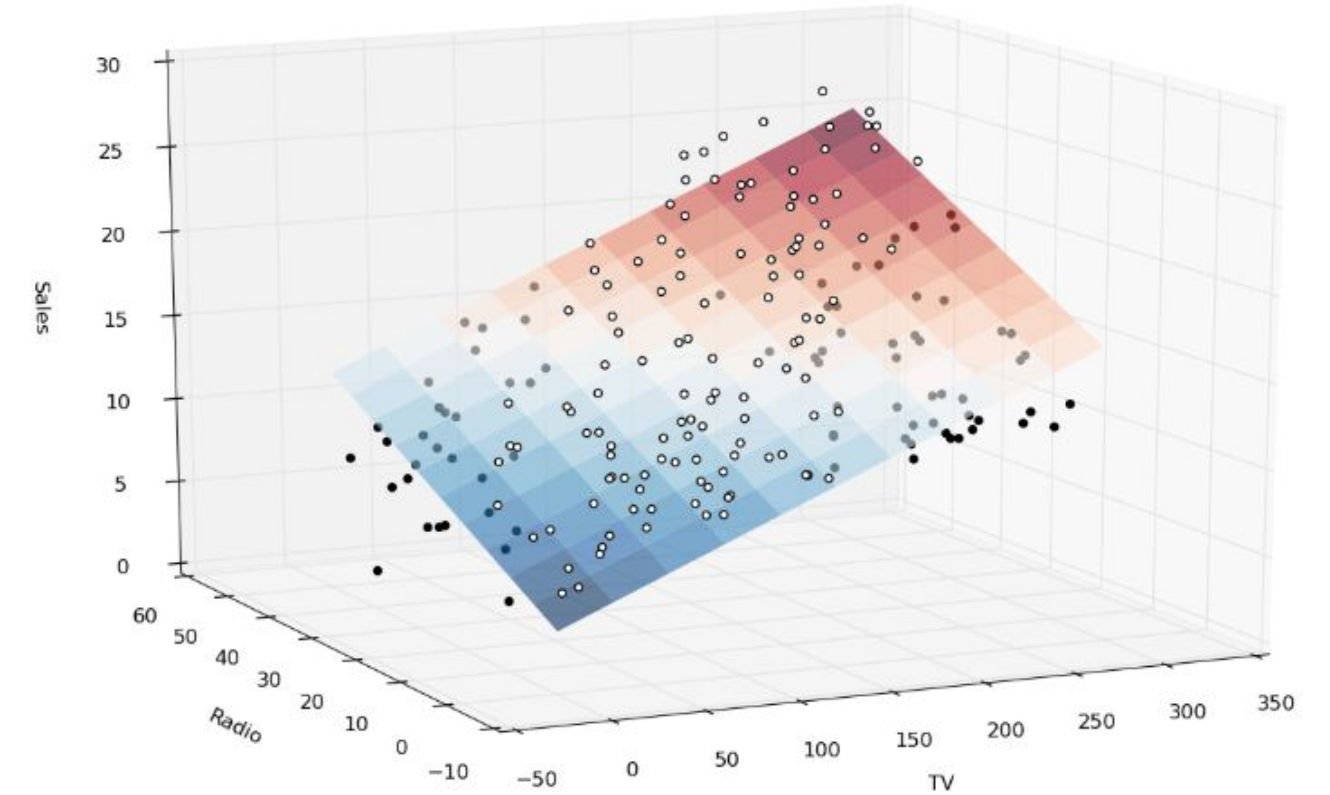
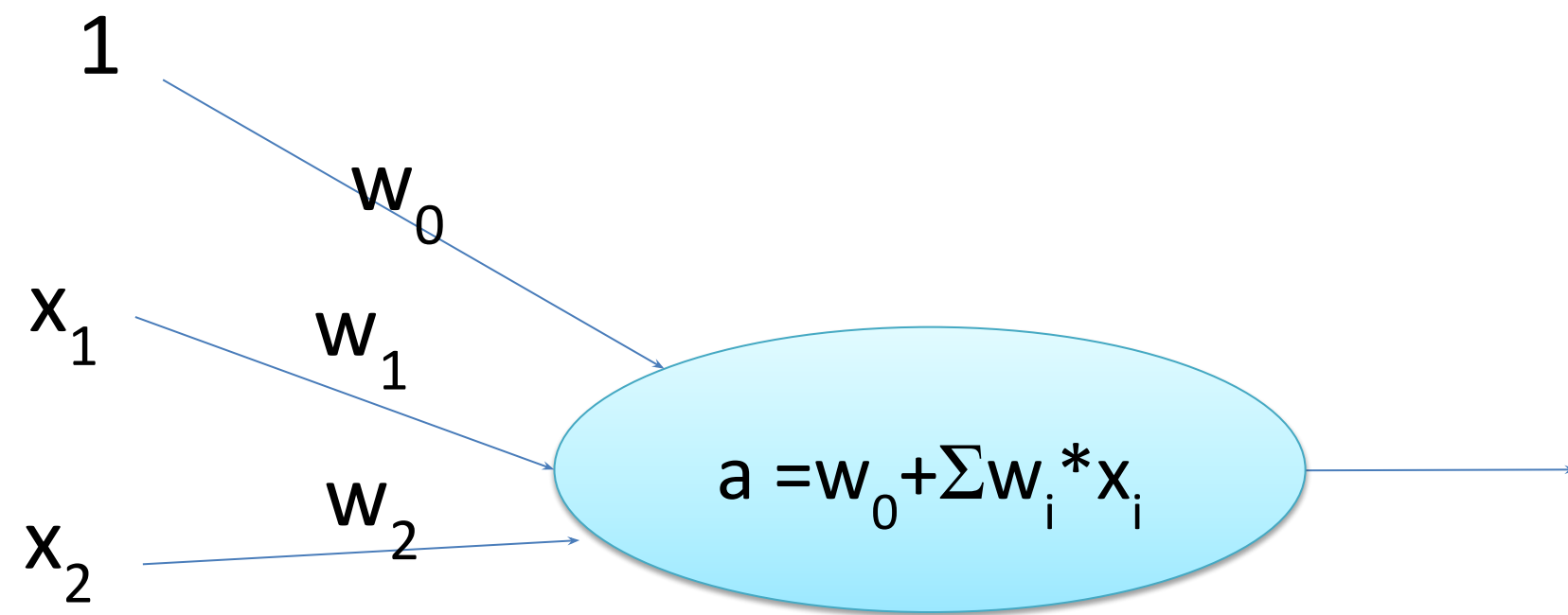
Neural Networks



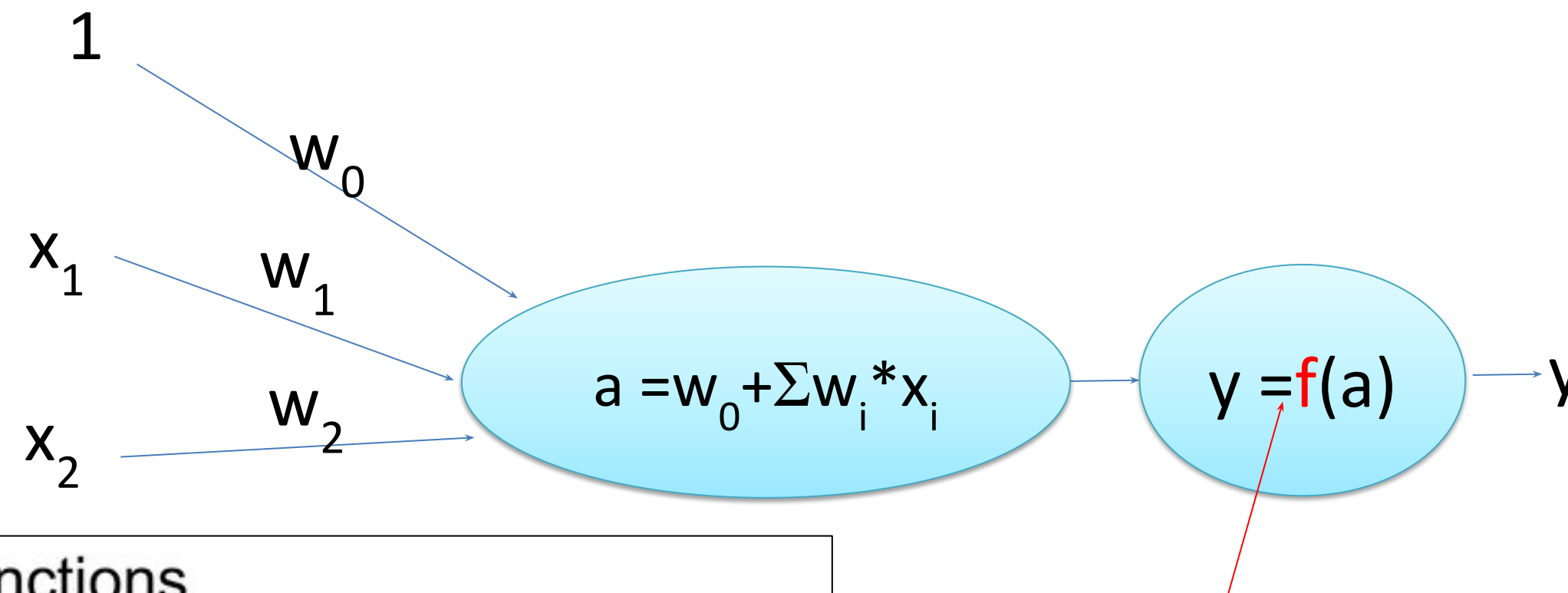
Neural Networks



Neural Networks



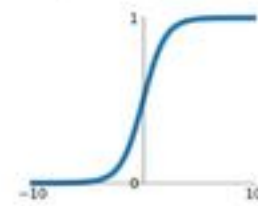
Neural Networks



Activation Functions

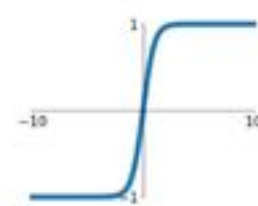
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



tanh

$$\tanh(x)$$



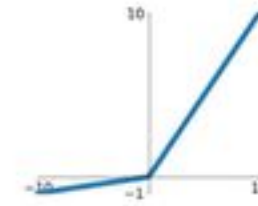
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

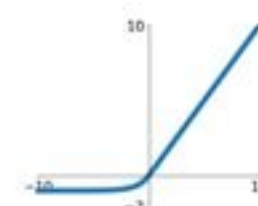


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

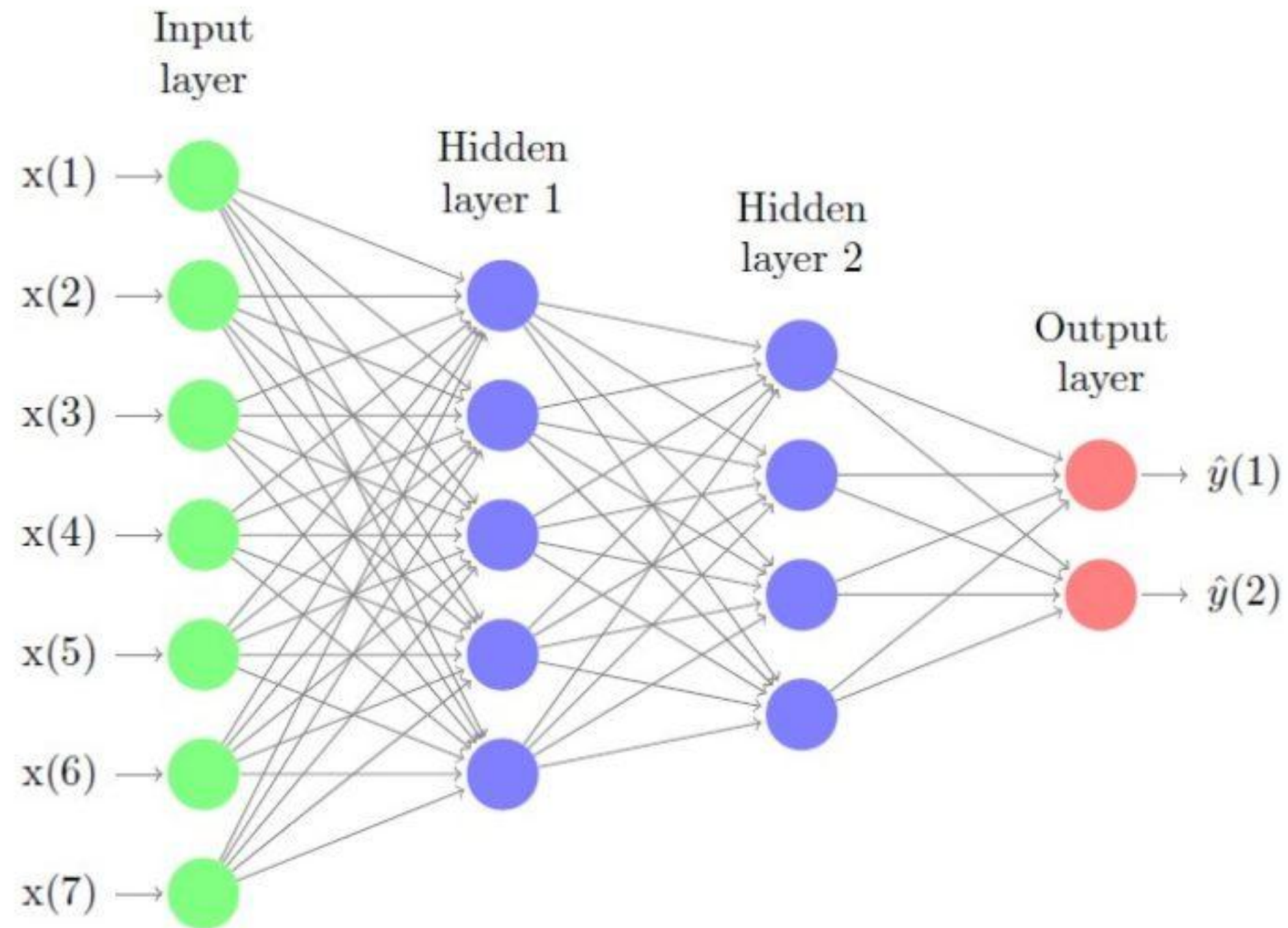
$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



**Activation
function**

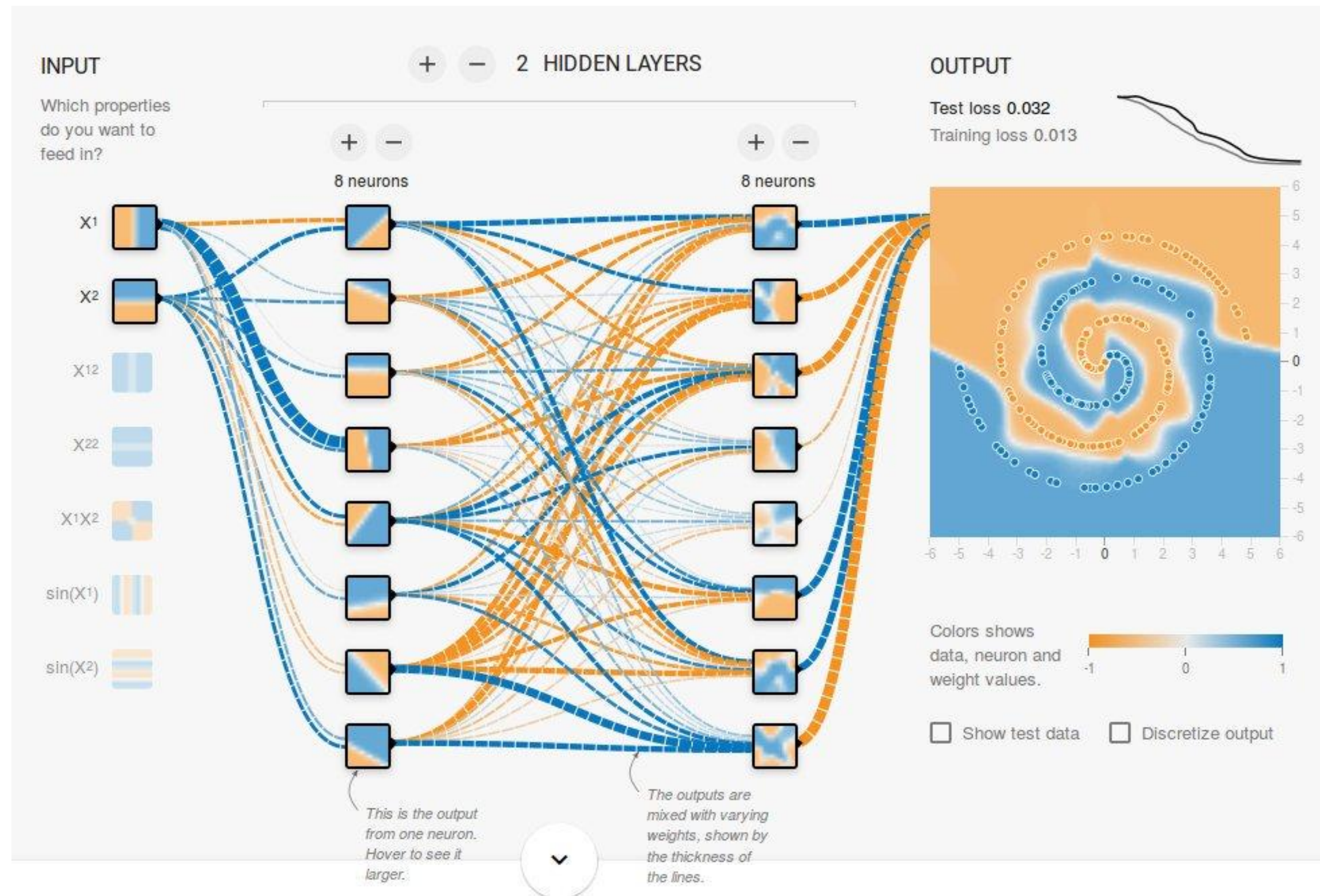
Typical NN

Multilayer perceptron



2D Feature Extraction Example

Simple case with fully connected NN



<https://www.coursera.org/learn/machine-learning>

Aurelien Geron. Hands-On Machine Learning with Scikit-Learn, Keras and Tensorflow.
Second edition. O'Reilly.

<https://jalammar.github.io/visual-interactive-guide-basics-neural-networks/>

<https://playground.tensorflow.org>



**Дякую
за увагу**

Контакти спікера

Email

yurii.chyrka@it-jim.com